

PTEN restrains SHH medulloblastoma growth through cell autonomous and nonautonomous mechanisms

Reviewed Preprint

v1 • November 11, 2025

Not revised

Zhimin Lao, Salsabiel El Nagar, Yinwen Liang, Daniel N Stephen, Alexandra L Joyner 

Developmental Biology Program, Sloan Kettering Institute, Memorial Sloan Kettering Cancer Center, New York, United States • Sophrosyne Bio, New York, United States • Biochemistry, Cell and Molecular Biology Program, Weill Cornell Graduate School of Medical Sciences, New York, United States

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eLife Assessment

This **valuable** study provides insights into the role of Pten mutations in SHH-medulloblastoma, by using mouse models to resolve the effects of heterozygous vs homozygous mutations on proliferation and cell death throughout tumorigenesis. The experiments presented are **convincing**, with rigorous quantifications and orthogonal experimentation provided throughout, and the models employing sporadic oncogene induction, rather than EGL-wide genetic modifications, represent an advancement in experimental design. However, the study remains limited, such that the biological conclusions do not extend greatly from those in the extant literature. This could be addressed with additional experimentation focused on cell cycle kinetic changes at early stages, as well as greater characterization of macrophage phenotypes (e.g., microglia vs circulating monocytes). The work will be of interest to medical biologists studying general cancer mechanisms, as the function of Pten may be similar across tumor types.

<https://doi.org/10.7554/eLife.108190.1.sa2>

Abstract

A third of patients with the pediatric cerebellar tumor Medulloblastoma (MB) have mutations that activate Sonic hedgehog (SHH) signaling (SHH-MB subgroup). The contribution of secondary mutations to tumor severity, however is not clear. *PTEN* mutations are enriched in the SHH-1 subtype that has the lowest survival rate. Widespread heterozygous loss of *Pten* in two SHH-MB mouse models increases penetrance and accelerates onset of differentiated tumors. We delineated cellular and transcriptional changes that accelerate tumor growth and cause differentiation using a sporadic SHH-MB mouse model expressing oncogenic *SmoM2* in rare cerebellar granule cell precursors (GCPs) and scRNA-seq analysis. Homozygous but not heterozygous sporadic loss of *Pten* resulted in rapid acceleration of tumor growth and end stage disease by 40 days, compared to ~25% survival in control *SmoM2* mice at 100 days. Heterozygous *PTEN* mutations therefore should negatively impact disease outcome primarily with germline mutations. Loss of *Pten* in normal or *SmoM2*-expressing GCPs increased proliferation and enhanced progenitor state initially but by 12 days *Pten* mutant *SmoM2*

tumors were highly differentiated due to increased survival of non-proliferating GCPs. Furthermore, macrophage infiltration and cytotoxicity were reduced in differentiated regions of tumors lacking *Pten*, indicating cell nonautonomous changes also contribute to accelerated tumor growth.

Introduction

Medulloblastoma (MB) is a brain tumor seen mainly in pediatric patients and which has four molecular subgroups, WNT, SHH, Group 3 and Group 4¹. The Sonic hedgehog subgroup of MB (SHH-MB) constitutes approximately 30% of MB and is characterized by mutations that constitutively activate SHH-signaling. Most SHH-MBs have one of four types of mutations that activate SHH-signaling: activating mutations in *SMO*, loss of function mutations in the pathway inhibitors *PTCH1* and *SUFU* or loss of *TP53* in association with amplification of the pathway effectors *GLI2* or *MYCN*. SHH-MB is further divided into four subtypes based on molecular, biological and clinical parameters^{2–6}. The work on SHH-MB subtypes has highlighted that particular secondary mutations likely contribute to tumor severity. For example, heterozygous *PTEN* and *KMT2D* loss-of-function mutations are enriched in the SHH-1 or beta subtype which has the lowest overall survival rate and highest frequency of metastasis and occurs primarily in infants (<4 years). Our recent work on the chromatin modifier *KMT2D* using two SHH-MB mouse models showed that when one or two copies of *Kmt2d* are deleted, not only is there a nearly two-fold increase in tumor penetrance with a Classic tumor cytoarchitecture (highly proliferative), but strikingly a near 100% penetrance of spinal cord metastasis and hindbrain invasion⁷. The transcriptional/chromatin landscape of mouse *Kmt2d* mutant SHH-MB primary and metastatic tumors was found to be shifted towards decreased expression of neural differentiation genes and an increase in oncogenes seen in more aggressive tumors compared to tumors with intact *Kmt2d*⁷. Thus, mouse models support the clinical data that loss of *KMT2D* contributes to more aggressive disease.

The gene encoding the dual specificity phosphatase *PTEN* (Phosphatase and TENsin homolog deleted on chromosome 10), is one of the most frequently mutated tumor suppressor genes in human cancers⁸, and is mutated in ~5% of SHH-MB³. *PTEN* regulates an array of cellular functions, including cell survival, proliferation and metabolism through attenuating phosphoinositide-3 kinase (PI3K-AKT) signaling and genome stability through nuclear functions. Recent work showed that *PTEN* also can be secreted and act in a cell nonautonomous manner^{9–11}. *PTEN* is secreted by tumor cells via binding to TMED10 and then binds to PLXDC2 on tumor associated macrophages (TAMs) and induces inflammatory status through triggering JAK2-STAT signaling, thus contributing to the tumor suppressor function of *PTEN*. Mouse models of SHH-MB indicate loss of *Pten* increases tumor incidence, but the underlying cell autonomous and nonautonomous reasons and impact on metastasis are not as clear^{12, 13}.

SHH-medulloblastoma arises in *Atoh1*-expressing granule cell precursors (GCPs) on the surface of the developing cerebellum. These precursor cells are dependent on SHH-signaling for their sustained proliferation and major expansion during the first two weeks after birth in mice^{14–16}. In a mouse model of SHH-MB in which a *NeuroD2* promoter drives expression of an oncogenic form of *SMO* (*SmoA1*) in the GCP-lineage, heterozygous loss of *Pten* in all cells (null mutation) or conditional knock-out in Nestin-expressing cells, which includes all cerebellar cells¹⁷, leads to increased penetrance and earlier onset of tumor formation with no increase in metastasis¹². Whereas most mouse models of SHH-MB have a classic (highly proliferative) histology, the *Pten* heterozygous mutant tumors have a nodular character with extensive regions of neural differentiation (NeuN+ cells) and an apparent decrease in cell death and increase in angiogenesis. In a mouse SHH-MB conditional model in which *Ptch1* is homozygously deleted at embryonic day 14 (E14) in all GCPs (*Atoh1-Cre*), additional homozygous loss of *Pten* leads to rapid tumor formation, and heterozygous loss leads to a slightly more aggressive phenotype than *Ptch1*

loss alone¹³. Similar to *NeuroD2*-SmoA1 tumors, in *Ptch1* mutant tumors *Pten* loss causes highly differentiated tumors with decreased cell death. In a mutagenic screen for increased SHH-MB penetrance and metastasis using a Sleeping beauty transposon approach in mice with heterozygous mutations in *Ptch1* or *Trp53*, insertions found in tumors with spinal cord metastasis were associated with mutations in *Pten*, *Akt2*, *Pik3R* and *Igf2*, implicating increased PI3K signaling in metastasis¹⁸. Consistent with this, in a mouse model where *Shh* is expressed in GCPs, additional expression of an activated form of AKT increases tumor incidence and spinal cord metastasis¹⁸. Data from human tumors support that a reduction in *PTEN* protein enhances tumor progression, since SHH-MB human patients with reduced or no *PTEN* protein have a worse overall survival¹². However, no correlation was seen between *PTEN* protein level and histology or metastasis. Thus, several questions remain as to whether *Pten* loss increases metastasis, what accounts for the faster growth and a more differentiated cellular phenotype in mouse *Pten* mutant SHH-MBs and why only in some models *Pten* heterozygous loss is sufficient to greatly enhance tumor progression.

We addressed these questions by comparing tumor progression and spinal cord metastasis in a sporadic mouse model of SHH-MB in which rare GCPs are induced to express an oncogenic form of SMO, SmoM2, at postnatal day 0 (P0) and have deletion of one or both copies of *Pten*. We found that only homozygous loss of *Pten* combined with SmoM2 expression enhances tumor progression in this sporadic model, and metastasis is not increased. Furthermore, the differentiated phenotype of the tumors is primarily due to a specific decrease in cell death of non-proliferative cells in the tumors, whereas cell death is as high in regions of high proliferation as in *SmoM2* tumors with normal *Pten* or heterozygous loss of *Pten*. Loss of *Pten* in normal GCPs, or those expressing SmoM2, leads to a mild initial increase in the proliferation index, indicating a cell autonomous function of *PTEN* in reducing proliferation. Single cell RNA-sequencing (scRNA-seq) analysis revealed that proliferating tumor cells that lack *Pten* not only have the expected increase in mTOR signaling, but also have altered expression of genes involved in neural differentiation and down regulation of inflammatory and interferon alpha response genes. Homozygous loss of *Pten* in GCPs also leads to a major decrease in infiltration of macrophages into tumors and an altered transcriptome indicating reduced cytotoxicity, revealing a possible cell nonautonomous function of *PTEN* that normally reduces tumor progression.

Results

***PTEN* mutations co-occur with mutations that activate SHH-signaling in human SHH-MB**

Using the cBioPortal^{20, 21} and a dataset of 127 patient samples considered to be SHH-MB³, we determined whether patients with mutations that activate SHH-signaling have *PTEN* mutations. 4.3% of the 127 samples had a mutation in *PTEN* and furthermore 4/5 of the mutations co-occurred with mutations that activate SHH-signaling (*Ptch1*, *SUFU* or *SMO*; Fig. 1A). Thus, although the *PTEN* mutations were detected in a low percentage of patient tumors, *PTEN* appears to be predominantly mutated in SHH-MB along with mutations that activate SHH-signaling.

Homozygous loss of *Pten* in sporadic SHH-MB accelerates growth and increases penetrance to 100%

We next used our mosaic mutant technique (MASTR)²² to test the impact of lowering *Pten* in a sporadic mouse model of SHH-MB. The approach involves inducing expression of an oncogenic form of the SHH receptor SMO (SmoM2) and eGFP in ~20,000 GCPs (~0.1%) at P0 (Fig. 1B).^{22–24} The model allows penetrance and survival time to be determined since less than 75% of mutant mice die by 100 days. Using the model, we asked whether loss of one or both copies of

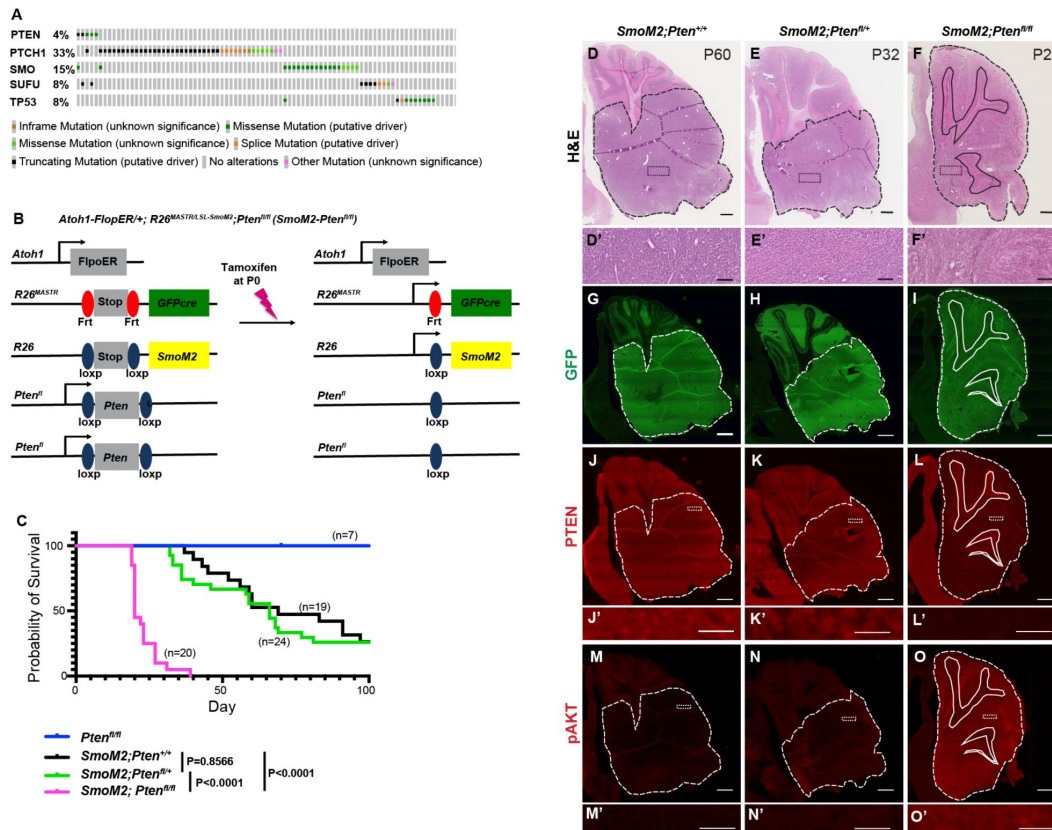


Figure 1.

Homozygous but not heterozygous loss of *Pten* in a sporadic mouse model of *SmoM2* SHH-MB increases penetrance and reduces time of disease onset

(A) An Oncoprint of 27 patient samples considered to be SHH-MB³ using the cBioPortal^{20, 21} showing the frequency of mutations in *PTEN* and genes that activate the SHH pathway showing co-occurrences. (B) Schematic showing Mosaic mutant Analysis with Spatial and Temporal Regulation (MASTR) technique used to model SHH-MB by expressing *SmoM2* in scattered GCPs starting at P0 and deleting one or two copies of *Pten* (*Atoh1-FlpoER/+; R26^{MASTR/LSL-SmoM2}; Pten^{fl/fl}* or *SmoM2-Pten^{fl/fl}* mice). (C) Kaplan-Meier curve and statistics (below) for survival of *SmoM2* mice compared to *SmoM2-Pten^{fl/fl}* and *SmoM2-Pten^{fl/fl}* mice. Statistics were calculated using the log-rank test. (D-F) Representative images of hematoxylin and eosin (H&E) stained sagittal sections of end stage tumors from *SmoM2*, *SmoM2-Pten^{fl/fl}*, and *SmoM2-Pten^{fl/fl}* mice with higher magnification images shown below (n>8 per genotype). Scale bars indicate 1mm (D-F) and 100 μ m (D'-F'). (G-I) GFP stained sagittal sections of end stage tumors from the indicated mutants (n=5-8 per genotype). Scale bar indicates 1mm. (J-L) Sagittal sections of end stage tumors from the indicated mutants stained for PTEN (J-L) or pAKT (M-O) with higher magnifications below. Scale bars indicate 1mm (J-O) and 20 μ m (J'-O'). (D-O) Dashed lines outline the tumors, solid lines outline IGL, dotted rectangles indicate location of high magnification images.

*Pten*²⁵ in rare GCPs-expressing *SmoM2* changes the survival curves of mice by comparing *Atoh1-FlpoER/+; R26^{MASTR/LSL-SmoM2}* mice (referred to as *SmoM2*) to *Atoh1-FlpoER/+; R26^{MASTR/LSL-SmoM2}; Pten^{fl/fl}* (referred to as *SmoM2-Pten^{fl/fl}*) and *Atoh1-FlpoER/+; R26^{MASTR/LSL-SmoM2}; Pten^{fl/+}* (referred to as *SmoM2-Pten^{fl/+}*) mice administered tamoxifen at P0 (Fig. 1B). Strikingly, all *SmoM2-Pten^{fl/fl}* mice succumbed to disease between P20 and P40, whereas *SmoM2* and *SmoM2-Pten^{fl/+}* mice began dying at ~P40 and by P100 ~25% of mice of both genotypes survived (Fig. 1C). Thus, loss of both *Pten* alleles is required to accelerate tumor progression in a mosaic *SmoM2* model of SHH-MB, unlike a *SmoA2* model with germline or broad deletion of *Pten* in the cerebellum where heterozygous deletion is sufficient¹².

Homozygous and heterozygous loss of *Pten* does not increase metastasis to the spinal cord in *SmoM2* SHH-MB

Given the different reported findings as to whether *Pten* loss increases metastasis in mouse models of SHH-MB, we asked whether in our sporadic *SmoM2* mice loss of *Pten* alters the frequency of spinal cord metastasis when mice show signs of brain tumor disease. Coronal spinal cord sections of each mouse (n=55-138 sections per mouse) were analyzed by cytology for the presence of spinal cord metastasis and tumors were confirmed with immunohistochemical (IHC) staining for GFP, Ki67 and PTEN (Figure 1-figure supplement 1A-C). Quantification of the percentage of mice that had at least 2 sections with more than a few tumor cells showed a similar tumor incidence between the three genotypes (*SmoM2*: 64%, n=11; *SmoM2-Pten^{fl/+}*: 69%, n=13; *SmoM2-Pten^{fl/fl}*: 58%, n=12). Furthermore, quantification of the percentage of spinal cord sections per mouse with tumor cells showed no difference between the *Pten* mutant tumors and *SmoM2* (Figure 1-figure supplement 1D). Likewise, measuring the area of the tumors did not detect a difference in metastatic disease when *Pten* was lowered or deleted (Figure 1-figure supplement 1E, F). Thus, in mouse sporadic *SmoM2* SHH-MB, heterozygous or homozygous loss of *Pten* does not appear to increase spinal cord metastasis.

Pten loss in a *SmoM2* sporadic mouse model of SHH-MB induces highly differentiated tumors

Hematoxylin and eosin (H&E) staining of primary tumor sections indicated that homozygous loss of *Pten* results in tumors with decreased cellularity (Fig 1 D-F). IHC staining of tumor sections at end stage confirmed that most tumor cells expressed GFP (Fig 1. G-I) and only *SmoM2-Pten^{fl/fl}* tumors lacked PTEN (Fig 1. J-L) and had strong Phospho-AKT (pAKT) (Fig 1. M-O) and pS6 staining (Figure 2-figure supplement 1A-D), thus confirming absence of PTEN and activation of mTOR (mammalian target of rapamycin) signaling only in *SmoM2-Pten^{fl/fl}* tumors. Staining with markers of cell proliferation (Ki67) and differentiation revealed that whereas *SmoM2* tumors consisted of mostly Ki67+ cells, *SmoM2-Pten^{fl/fl}* tumors were highly differentiated expressing the post mitotic marker P27 and neural differentiation marker NeuN with only small areas of proliferative cells (Fig. 2A-H). The *SmoM2* tumors had broad expression of the SHH target gene *Gli1*, whereas the *SmoM2-Pten^{fl/fl}* tumors expressed *Gli1* mRNA only in Ki67+ regions of the tumors (Fig. 2I-L), which was complementary to the neural differentiation marker Synaptophysin (SYP/Syp) that was absent from *SmoM2* tumors (Fig. 2K-P). Thus, despite *SmoM2* being expressed in the differentiated cells of the *SmoM2-Pten^{fl/fl}* tumors, SHH-signaling is down regulated specifically in differentiated cells. Interestingly, in 3/9 tumors from *SmoM2-Pten^{fl/+}* mice some sections had patches of cells devoid of PTEN and positive for pAKT, indicating spontaneous loss of the second allele of *Pten* in rare cells that expand as a coherent group (Figure 2-figure supplement 1E-H). The regions without PTEN in *SmoM2-Pten^{fl/+}* mice had a similar differentiated phenotype (SYP+ and NeuN+) to *SmoM2-Pten^{fl/fl}* tumors (Figure 2-figure supplement 1I-K).

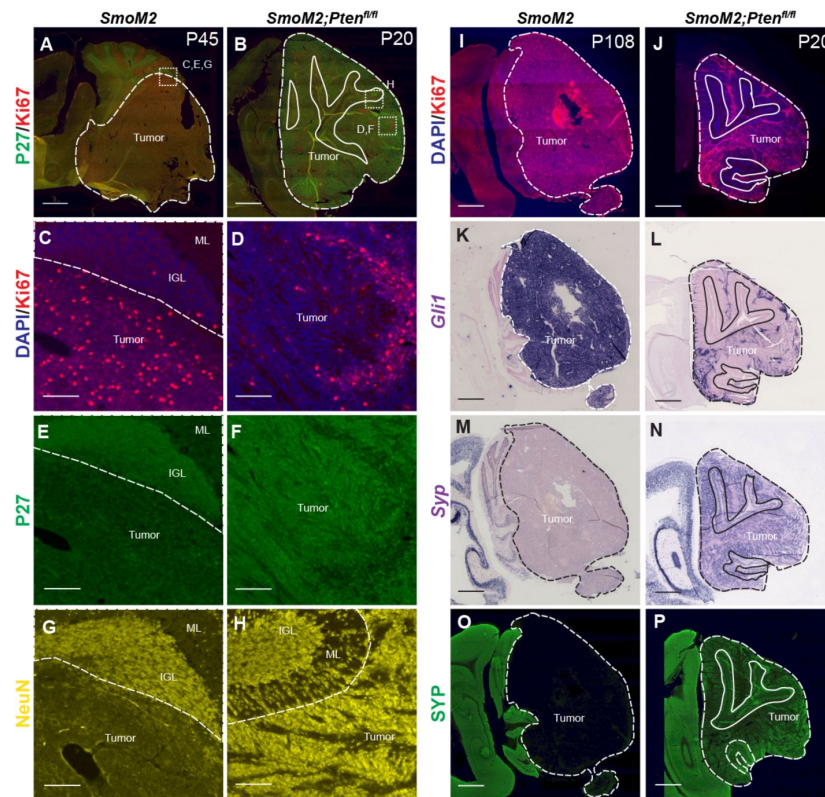


Figure 2.

***Pten* loss in *SmoM2* sporadic SHH-MB induces highly differentiated tumors**

(**A, B**) Sagittal sections of the cerebellum showing end stage tumors from *SmoM2* and *SmoM2-Pten^{fl/fl}* mice stained for P27 (post mitotic cells) and Ki67 (proliferating cells) (*SmoM2*, n=5; *SmoM2-Pten^{fl/fl}* n=7). Dashed lines outline each tumor, solid lines outline IGL, dotted squared indicate location of high magnification images in (C-H). (**C-H**) High power images of sections stained for Ki67 and DAPI (C, D), P27 (E, F) or NeuN (G, H) (*SmoM2*, n=5; *SmoM2-Pten^{fl/fl}* n=7). Dashed line separates the internal granule layer (IGL) or molecular layer (ML) from the tumor. (**I, J**) Sagittal sections of the cerebellum showing lateral tumors from *SmoM2* and *SmoM2-Pten^{fl/fl}* mice stained for Ki67 and DAPI (*SmoM2*, n=7; *SmoM2-Pten^{fl/fl}* n=8). (**K-N**) RNA *in situ* hybridization staining of sections for *Gli1* (K, L) and *Syp* mRNA (M, N) (n=3 per genotype). (**O, P**) Immunohistochemical staining of sections for SYP protein (n=6 per genotype). (I-P) Dashed lines outline the tumors, solid lines outline IGL. (A, B, I-P) Scale bars indicate 1mm. (C-H) Scale bars indicates 100µm.

***Pten* loss protects differentiated tumor cells from cell death**

Mouse SHH-MB models that have a highly differentiated cytology are rare and have primarily been reported in two *Pten* models.^{12,13} In a *Ptch1* mutant model of SHH-MB, it was shown that proliferating tumor cells give rise to NeuN+ postmitotic tumor cells that then die at a high rate²⁶. We therefore asked whether *Pten* loss (increased mTOR signaling) rescues the post-mitotic tumor cells from death by staining for SYP or NeuN and TUNEL (**Fig. 3A-H**). *SmoM2-Pten^{fl/fl}* tumors had significantly less TUNEL particles in regions of high SYP labeling (SYP Hi) compared to SYP low regions (**Fig. 3C-I**, **Figure 3-figure supplement 1**). Moreover, in the regions of low SYP, the density of TUNEL staining was similar to in *SmoM2* tumors where SYP is low throughout the tumor. (**Fig. 3J**). As expected, in PTEN negative patches of *SmoM2-Pten^{fl/+}* tumors, TUNEL staining was lower in these SYP Hi regions of differentiated cells (NeuN+) than in SYP low proliferative regions (Ki67+) (**Fig. 3J,K** and **Figure 2-figure supplement 1I-L**).

Loss of *Pten* leads to rapid cell autonomous changes in *SmoM2* GCP behaviors

Given how rapidly the tumors grow in the *SmoM2-Pten^{fl/fl}* mice and that lethal disease occurs between P20-40, we asked how early a differential growth phenotype can be seen in the three genotypes. In our previous study of the effect of loss of *Kmt2d* in SHH-MB models, no difference in tumor size was detected at P12 but was by P21⁷. We therefore asked whether *SmoM2-Pten^{fl/fl}* tumors have a growth advantage over *SmoM2* tumors at P12. Staining of sagittal sections of *SmoM2* tumors of the three *Pten* genotypes with EdU (1 hour pulse) and GFP to mark the proliferating GCPs in the external granule cell layer (EGL) showed a massive expansion of the EGL in *SmoM2-Pten^{fl/fl}* mice compared to *SmoM2* mice (**Fig. 4A-C**). Quantification of cerebellar area in midline sections revealed a significant increase in *SmoM2-Pten^{fl/fl}* mice compared to *SmoM2* mice and *SmoM2-Pten^{fl/+}* mice (**Fig. 4D**). The EGL area alone showed a 4-fold average area increase in *SmoM2-Pten^{fl/fl}* mice compared to *SmoM2* mice and *SmoM2-Pten^{fl/+}* mice had a slight increase (<2-fold) compared to *SmoM2* mice (**Fig. 4E**). As in end stage tumors, the tumors in *SmoM2-Pten^{fl/fl}* mice had a significant decrease in dying cells compared to the other two genotypes at P12 (**Fig. 4F-L**, **Fig. 4-figure supplement 1**). In large lesions with extensive differentiation (NeuN+ Ki67-cells) the density of TUNEL particles was decrease in NeuN+ regions compared to proliferative regions. As expected, pAKT staining was upregulated in the EGL of *SmoM2-Pten^{fl/fl}* mice with no detectable staining in the other two genotypes (**Fig. 4M-O**).

One prediction of our findings is that once *SmoM2* GCPs lacking *Pten* differentiate their survival should be enhanced. Therefore, if proliferating GCPs are labeled and then fate mapped for two days, a higher percentage of *SmoM2-Pten^{fl/fl}* labeled cells should be differentiated (post-mitotic) compared to in *SmoM2* mice. To test this idea, we administered one injection of BrdU into P10 mice and then sacrificed them at P12 and quantified the percentage of BrdU+ cells in the EGL that were Ki67- (**Fig. 4P**). As predicted, a higher percentage of BrdU+ cells in *SmoM2-Pten^{fl/fl}* mice were Ki67- at P12 compared to *SmoM2* (**Fig. 4Q**). This result indicates that by P10 after loss of *Pten* at P0, either differentiation is promoted in *SmoM2* GCPs and/or that the differentiated progeny of GCPs lacking PTEN are spared from cell death. The proliferation rate of GCPs in *SmoM2* and *SmoM2-Pten^{fl/fl}* mice were similar at P12 (**Fig. 4R**), as determined by injecting EdU 1 hour before sacrificing the mice (%EdU+ Ki67+/Ki67+ cells), indicating the latter possibility.

Given that at P12 the cellular phenotype of GCPs in *SmoM2-Pten^{fl/fl}* mice was similar to in mice at end stage, we analyzed GCPs at P8, eight days after the mice were treated with tamoxifen. Similar to at P12, the area of the proliferating EGL of the cerebellum was significantly greater in both the *SmoM2-Pten^{fl/fl}* and the *SmoM2-Pten^{fl/+}* mice compared to *SmoM2*, although the area of the cerebellum was similar across genotypes likely due to the relatively small increase in EGL area

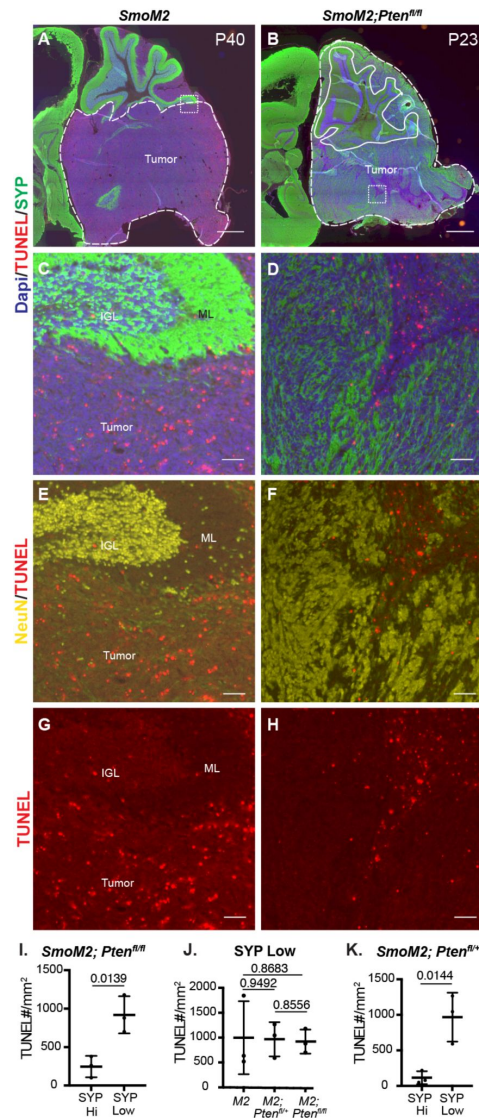


Figure 3.

***Pten* loss in *SmoM2* SHH-MB protects differentiated tumor cells from cell death**

(A, B) Sagittal sections of end stage tumors in *SmoM2* and *SmoM2-Pten^{fl/fl}* mice stained as indicated (*SmoM2*, n=3; *SmoM2-Pten^{fl/fl}* n=4). Dashed lines outline the tumors, solid lines outline IGL. Dotted line square indicates location of high magnification images show in (C-H). Scale bars indicate 1mm. (C-H) High magnification images of tumors stained as indicated showing reduced cell death (TUNEL particles) in differentiated regions (NeuN+ and SYP+) of tumors in *SmoM2-Pten^{fl/fl}* mice compared to *SmoM2* (*SmoM2*, n=3; *SmoM2-Pten^{fl/fl}* n=4). Scale bars indicates 50μm. (I-K) Quantification of density of TUNEL particles in tumor regions with high SYP (SYP HI) compared to low (SYP Low) staining in *SmoM2-Pten^{fl/fl}* mice (I), in SYP Low regions in *SmoM2* (M2), *SmoM2-Pten^{fl/fl}* (M2; *Pten^{fl/fl}*) and *SmoM2-Pten^{fl/fl}* (M2; *Pten^{fl/fl}*) mice (J), and in SYP High and Low regions of *SmoM2-Pten^{fl/fl}* mice (K). n=3 mice/genotype. All statistics were determined using an unpaired t-test comparing SYP High to Low (I, K) or two pairs of genotypes (J).

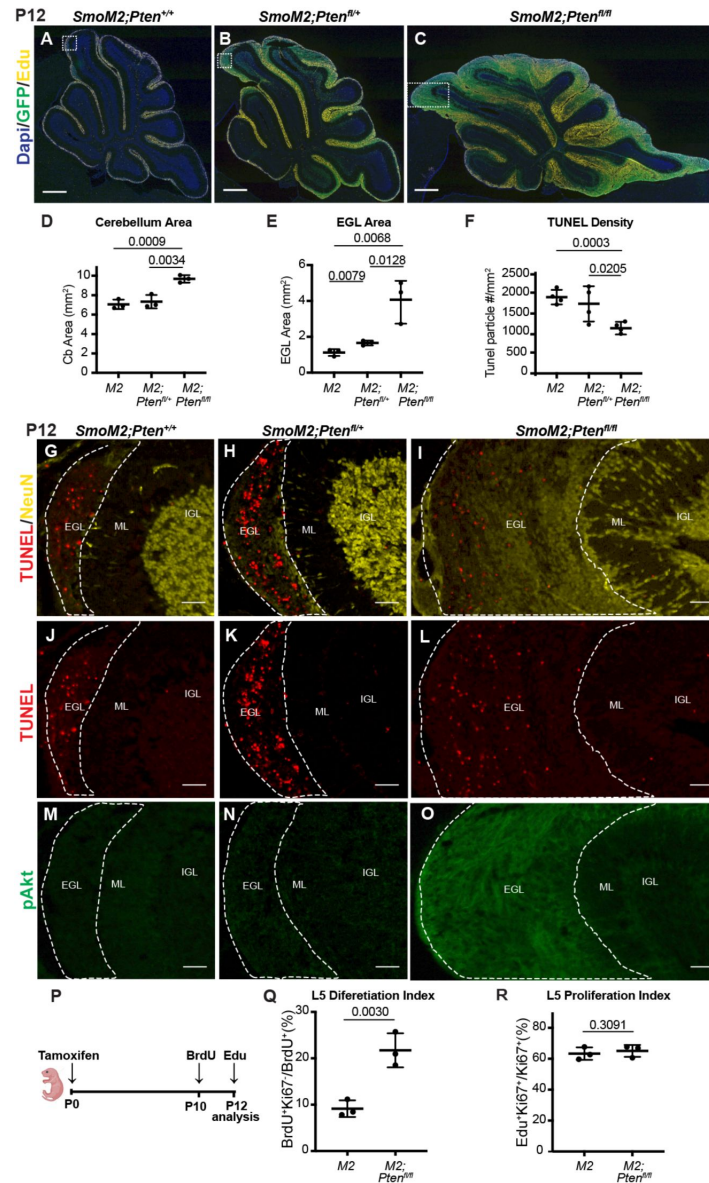


Figure 4.

Loss of *Pten* leads to cell autonomous changes in *SmoM2* GCPs at P12

(A-C) Midline cerebellar sagittal sections of P12 *SmoM2*, *SmoM2-Pten^{fl/+}* and *SmoM2-Pten^{fl/fl}* mice stained for GFP and Edu to show the EGL with proliferating mutant GCPs (*SmoM2*, n=11; *SmoM2-Pten^{fl/+}* n=5; *SmoM2-Pten^{fl/fl}* n=6). Scale bars indicate 0.5 mm. Dotted line boxes indicate locations of images in (A for G, J, M; B for H, K, N; C for I, L, O). (D) Quantification of the area of the cerebellum on midline sections of *SmoM2* (M2), *SmoM2-Pten^{fl/+}* (M2; *Pten^{fl/+}*) and *SmoM2-Pten^{fl/fl}* (M2; *Pten^{fl/fl}*) mice (n=3 mice/genotype). (E) Quantification of the area of the EGL on midline sections in the three genotypes (n=3 mice/genotype). (F) Quantification of TUNEL particle density (cell death) in the EGL of midline sections in the three genotypes (n=3 mice/genotype). (G-O) Staining of sections of the three genotypes for the proteins indicated showing reduced cell death (TUNEL density) and increased pAkt and NeuN in the EGL of *SmoM2-Pten^{fl/fl}* mice (n>3 per genotype). Scale bars indicates 50µm. EGL, external granule layer; ML, molecular layer (region between IGL and EGL); IGL, internal granule layer. (P) Schematic of experimental design. (Q) Quantification of the percentage of proliferating GCPs that became post mitotic (Ki67-) between BrdU labeling at P10 and analysis at P12 in a region of lobule 5 (L5) (n=3 mice/genotype). (R) Quantification of the percentage of GCPs that are in S phase (Edu+ Ki67+ of all Ki67+ GCPs) at P12 (n=3 mice/genotype). All statistics were determined using an unpaired t-test comparing two pairs of genotypes.

(~2-fold)(**Fig. 5A-E**). Quantification of the proliferation rate (%GFP+ EdU+/GFP+ cells in the outer EGL (oEGL) of lobule 4/5 (L5) containing proliferating cells was slightly but significantly higher in the *SmoM2-Pten^{fl/fl}* and the *SmoM2-Pten^{fl/+}* mice compared to *SmoM2* (**Fig. 5F**). Moreover, the percentage of mutant (GFP+) cells that remained in the outer EGL of all the GFP+ cells in L5 (including those that differentiated) was significantly higher in the two *Pten* mutant genotypes (**Fig. 5G**). This result indicates that loss or reduction of *Pten* initially induces a pro-proliferative or progenitor state in *SmoM2*-expressing GCPs. By P12 however, *SmoM2-Pten^{fl/fl}* GCP-lineage cells accumulate more in the EGL than do *SmoM2-Pten^{fl/+}* or *SmoM2* GCPs, in part because there is an accumulation of NeuN+/SYP+ granule cell (GC)-like cells, likely due to a survival advantage of these cells. This survival advantage suggests that the rapid growth of tumors in *SmoM2-Pten^{fl/fl}* mice is due to accumulation of GC-like cells and would account for the highly differentiated cytology of end stage tumors.

Loss of *Pten* in normal GCPs causes increased proliferation and decreased differentiation

Given that *Pten* loss initially increases proliferation and a progenitor state of *SmoM2* (SHH activated) GCPs, we asked whether *Pten* loss in normal GCPs has the same effect. Using our mosaic MASTR approach, we used *Atoh1-FlpoER* to induce GFPcre in scattered GCPs at P0, with or without concomitant deletion of both floxed copies of *Pten* (**Fig. 6A**). As in *SmoM2* GCPs, we found that at P8 a small but significant increase in the proliferation rate and in the proportion of GFP+ cells that remained in the proliferative outer EGL of *Pten* mutant GCPs (**Fig. 6B-E**). Using *Atoh1-Cre* to delete *Pten* in all GCPs in the embryo (E14 onwards), we found that the cerebellar area of *Atoh1-Pten* conditional mutants (*Atoh1/+; Pten^{fl/fl}*) was significantly increased at P5 and later, but not at P1 compared to *Atoh1/+; Pten^{fl/+}* control mice (**Fig. 6F-N**). Thus, PTEN normally constrains GCP proliferation and possibly differentiation.

Loss of *Pten* in *SmoM2* SHH-MB tumor cells results in a cell nonautonomous decrease in macrophage infiltration

Given the low density of dying cells in *SmoM2-Pten^{fl/fl}* SHH-MB, we asked whether the tumors have a change in the density of infiltrating macrophages. We previously showed that tumors in *SmoM2* mice have a high infiltration of macrophages and that they are pro-tumorigenic in a majority of tumors, because reduction of macrophages using PLX5622 treatment leads to a decrease in tumor incidence²⁷. Staining of cerebellar tumor sections from *SmoM2-Pten^{fl/fl}* and *SmoM2* mice at end stage revealed the expected high infiltration of IBA1+ macrophages in *SmoM2* tumors, but also a striking decrease in macrophage density in tumors lacking *Pten* (**Fig. 7A-D**). Moreover, co-staining with Ki67 or NeuN showed that in *SmoM2-Pten^{fl/fl}* mice the macrophages were preferentially located in the areas with proliferating cells and thus increased cell death (**Fig. 7C-F**). The decrease in macrophages in the differentiated regions of *SmoM2-Pten^{fl/fl}* mice is not due to a lack of blood vessels, based on CD31 staining of tumors (**Fig. 7G, H**). At P12, in small lesions the macrophages mainly accumulated in the meninges between the lobules but in large lesions they were more concentrated in regions of proliferation in both genotypes and as in end stage were decreased in NeuN+ regions in *SmoM2-Pten^{fl/fl}* mice (**Fig. 7I-P**). Thus, loss of *Pten* specifically in tumor cells leads to a cell nonautonomous decrease in macrophages in the tumor microenvironment from an early stage.

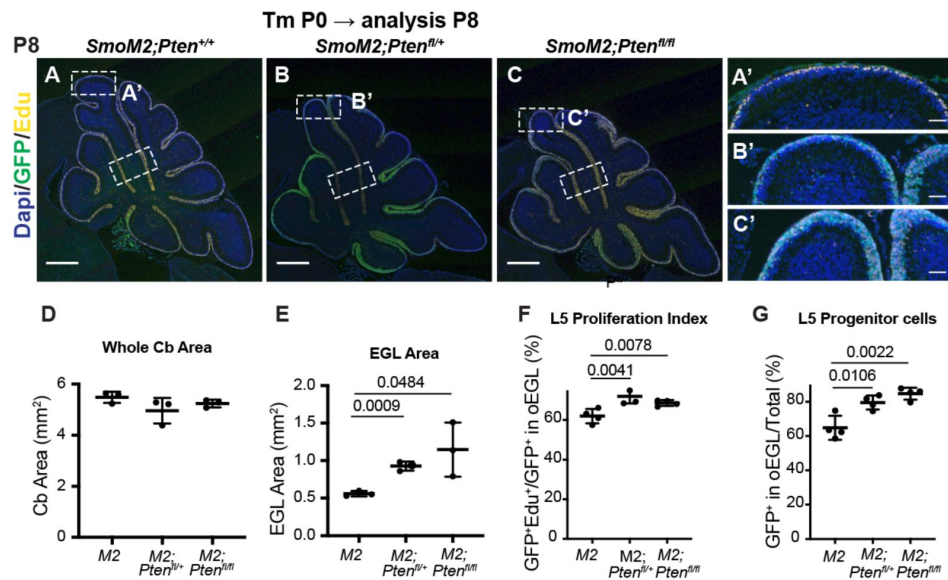


Figure 5.

Loss of *Pten* leads to cell autonomous changes *SmoM2* GCP behaviors at P8

(A-C) Midline cerebellar sagittal sections of P8 *SmoM2*, *SmoM2-Pten^{fl/+}* and *SmoM2-Pten^{fl/fl}* mice stained for GFP and EdU to show the EGL with proliferating mutant GCPs. Dotted line boxes indicate locations of where quantifications were performed. Scale bars indicate 0.5 mm. (A'-C') High magnification images of areas indicated in (A-C). Scale bars indicate 0.5 μm. (D) Quantification of the area of the cerebellum on midline sections of *SmoM2* (M2), *SmoM2-Pten^{fl/+}* (M2; *Pten^{fl/+}*) and *SmoM2-Pten^{fl/fl}* (M2; *Pten^{fl/fl}*) mice (n=3 mice/genotype). (E) Quantification of the area of the external granule layer (EGL) on midline sections in the three genotypes (n=3 mice/genotype). (F) Quantification of the proliferation index in lobule 5 (L5) at P8: percentage of mutant GCPs that are in S phase (GFP+ EdU+ of all GFP+ GCPs), (n=4 mice/genotype). (G) Quantification of the percentage of mutant GCPs that remained in the proliferative outer EGL (oEGL) at P8 and thus were progenitors. Analysis performed in wall of lobule 5 (lower rectangles in (A-C)) (n=4 mice/genotype). All statistics were determined using an unpaired t-test comparing two pairs of genotypes.

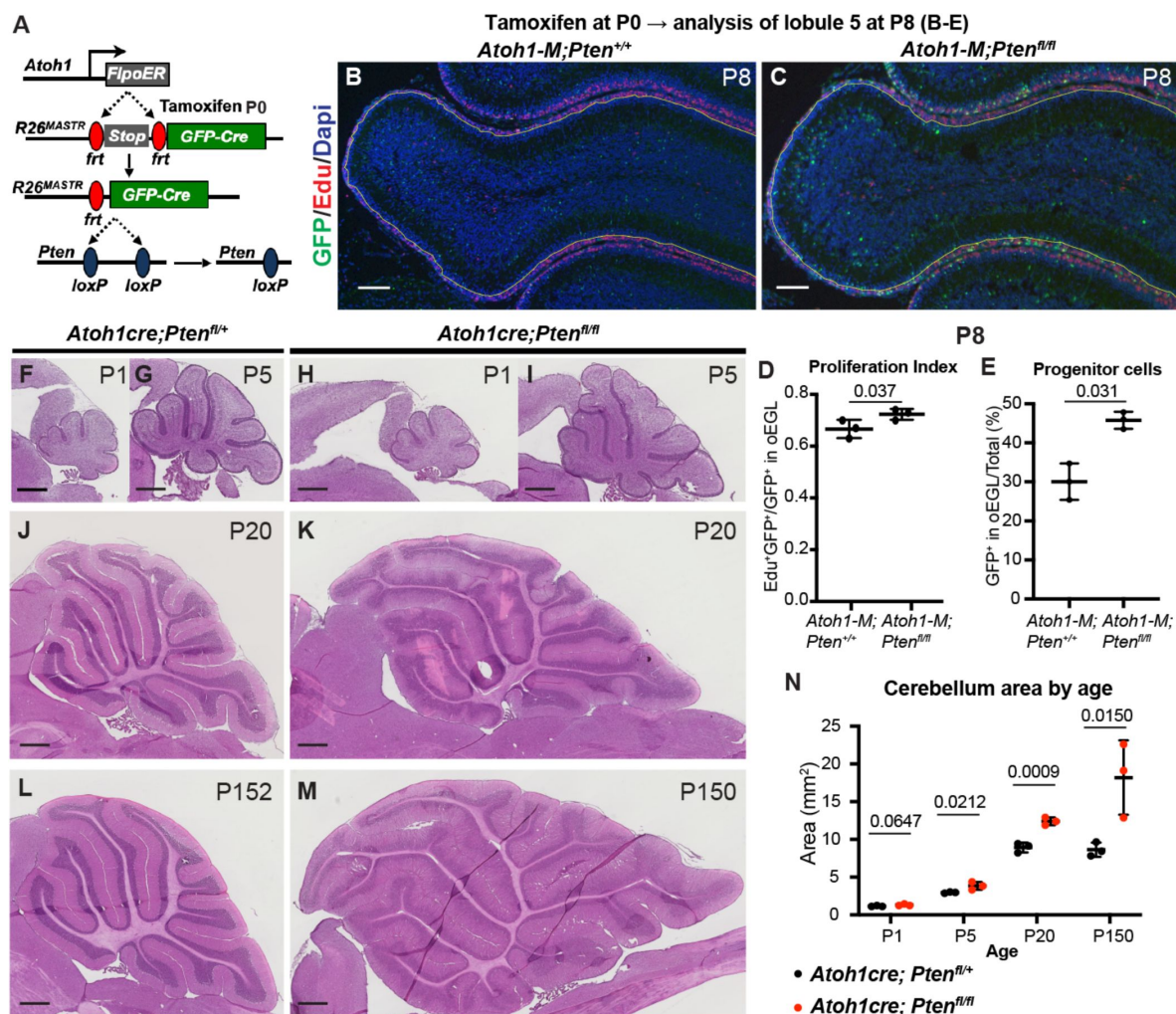


Figure 6.

Loss of *Pten* in normal GCPs causes increased proliferation and decreased differentiation

(A) Schematic showing mosaic mutant analysis approach to study scattered GFP+ *Pten* homozygous mutant GCPs in *Atoh1-FlpoER*; *R26^{FSF-GFPcre}*; *Pten^{fl/fl}* (*Atoh1-M-Pten^{fl/fl}*) mice compared to control GFP+ cells in *Atoh1-FlpoER*; *R26^{FSF-GFPcre}*; *Pten^{+/+}* mice (*Atoh1-M-Pten^{+/+}*) at P8 following tamoxifen at P0. (B, C) Midline sagittal sections of lobule 5 stained with GFP marking mutant cells and EdU (S phase) at P8 in the two genotypes. Scale bars indicate 100µm. (D) Quantification of the proliferation index in lobule 5 (L5) at P8: percentage of mutant GCPs that are in S phase (GFP+ EdU+ of all GFP+ GCPs; n=3 mice per genotype). (E) Quantification of the percentage of mutant GFP+ GCPs that remained in the proliferative outer EGL (oEGL) at P8 and thus were progenitors (n=3 mice per genotype). (F-M) Midline sagittal H&E stained sections of the cerebellum of conditional mutant mice lacking *Pten* in the GCP lineage (*Atoh1-Cre*; *Pten^{fl/fl}*) compared to controls (*Atoh1-Cre*; *Pten^{fl/+}*). Scale bars indicate 500µm. (N) Quantification of the area of midline sagittal sections of the cerebellum across development and aging (n=3 mice per age). All statistics were determined using an unpaired t-test comparing two pairs of genotypes (D, E) or two genotypes at each age (N).

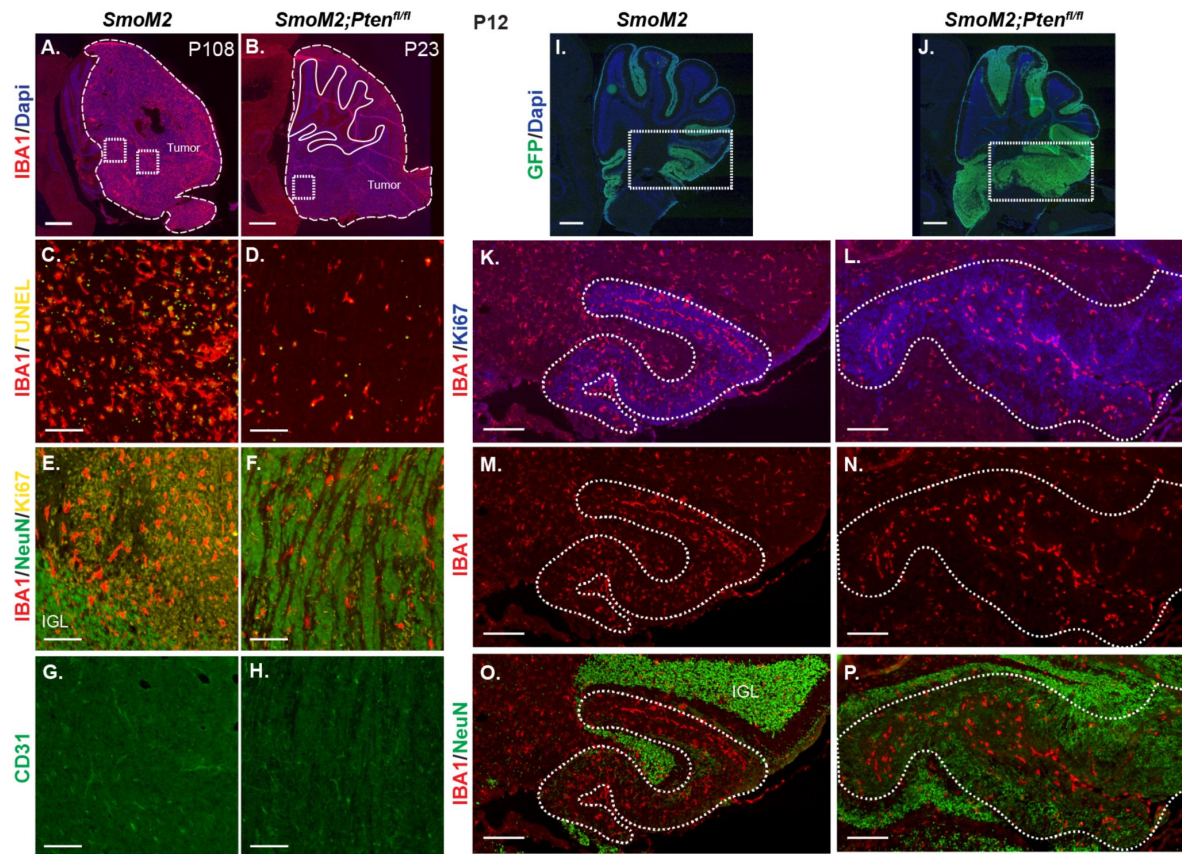


Figure 7.

Loss of *Pten* in *SmoM2* GCPs results in a cell non-autonomous decrease in macrophages in SHH-MB tumors

(A, B) Sagittal sections of end stage tumors in *SmoM2* and *SmoM2-Pten^{fl/fl}* mice stained with IBA1 (macrophage marker) and DAPI (n=3 mice/genotype). Dashed lines outline the tumors, solid lines outline IGL. Dotted line squares indicate locations of high magnification images show in (C-H). Scale bars indicate 1mm. (C-H) Higher magnification images of tumors stained with IAB1 and TUNEL (C, D), IAB1, NeuN and Ki67 showing enrichment of macrophages in undifferentiated regions of *SmoM2-Pten^{fl/fl}* mice (E, F) and with CD31 to label blood vessels (G, H) (n=5-9 mice/genotype). IGL, internal granule layer. Scale bars indicates 100µm. (I, J) Lateral sagittal cerebellar sections of P12 *SmoM2* and *SmoM2-Pten^{fl/fl}* mice stained for GFP and DAPI to show the EGL with mutant GCPs (n=3 mice/genotype). Dotted line boxes indicate locations of higher magnifications images in (K-P). Scale bars indicate 500µm. (K-P) Sections of P12 mice stained with IAB1, Ki67 and NeuN showing macrophages in large tumor lesions are enriched in proliferative regions of *SmoM2-Pten^{fl/fl}* mice (n=3 mice/genotype). Dotted line outlines the tumor lesion. Scale bars indicate 200µm.

scRNA-seq analysis reveals altered neural differentiation and suppression of immune signaling in tumor cells lacking *Pten*

To gain insight into transcriptional changes that occur in *SmoM2* SHH-MB tumor cells that lack *Pten* and that could account for the cell autonomous changes seen in the tumors, we performed single cell scRNA-seq of whole tumors dissected from *SmoM2* and *SmoM2-Pten^{fl/fl}* mice ($n = 2$ per genotype). We chose to analyze SHH-MBs from P16 *SmoM2-Pten^{fl/fl}* mice, since this is a few days before mice show signs of disease, and waited until P22 for *SmoM2* mice when the tumors are more visible. Marker analysis of sections of the tumors at the two ages confirmed that the cellular phenotypes of large tumors in *SmoM2-Pten^{fl/fl}* and *SmoM2* mice were similar to tumors at end stage (**Fig. 8-figure supplement 1**). The *SmoM2-Pten^{fl/fl}* tumors had extensive regions of neural differentiation that had reduced cell death (TUNEL particle density) and macrophage density (IBA1+ cells) compared to in proliferative regions (Ki67+) and in *SmoM2* tumors at P22 or P16.

Tumor cells from the two genotypes separated well in a PCA plot after filtering out poor-quality cells and integrating the replicates (**Fig. 8-figure supplement 2A-C**). Clustering of all cells (17,998 *SmoM2* and 22,864 *SmoM2-Pten^{fl/fl}*) was performed²⁸ to produce twenty clusters with most of the cells, as expected, expressing the GCP lineage marker *Pax6* (clusters 0-9 and 13), with nine additional small clusters that included astrocytes, oligodendrocytes, unipolar brush cells, macrophages, endothelial cells and mesenchymal cells (pericytes and fibroblast-like cells) based on marker gene expression (**Fig. 8A, B**, **Fig. 8-figure supplement 2D, E**, **Table S1**). The cells from each sample were distributed in all clusters, although the proportions of cells from each genotype varied between clusters (**Fig. 8C**, **Fig. 8-figure supplement 2F, G**, **Table S2**). Within the *Pax6*+ clusters 0-9, based on marker gene expression and cell cycle phase (**Fig. 8D, E**), clusters 3, 4, 8 were proliferative (Ki67+) and enriched for the GCP markers *Gli1*, *Atoh1* and *Barhl1*, indicating highly proliferative GCP-like tumor cells. Cluster 6 had a mixture of cells in G2/M and G1, indicating cells transitioning from a proliferative to differentiated state. Clusters 1, 5, 7 had post mitotic (G1) cells that expressed early GC differentiation markers including *Barhl1* and *Slc17a6*, indicating early GC-like cells. Clusters 0, 2, 9 were also in G1 but expressed the mature GC markers *Slc17a7* and *Syp* (**Fig. 8E**) and had a low proportion of cells expressing *Gfp* (**Fig. 8E**) and a much higher proportion of cells from *SmoM2* mice than from *SmoM2-Pten^{fl/fl}* (**Fig. 8C, F**, **Fig. 8-figure supplement 2G**) indicating mature GC-like cells. Many of the GC-like cells (clusters 0, 2, 9) were likely wild type GCs from the IGL that would disproportionately contaminate the samples from *SmoM2* mice as the tumors were smaller than in *SmoM2-Pten^{fl/fl}* mice. When only the GC-lineage clusters with extensive *Gfp* expression were considered (1, 3, 4, 5, 6, 7, 8), the proportions of cells from the two genotypes were fairly evenly distributed in all clusters (**Fig. 8G**).

Pseudotime analysis^{29, 30} reflecting the differentiation pathway of cells within clusters 0-9 with an origin in the GCP-like clusters was consistent with our cell type predictions, showing clusters 0, 2, 9 are most mature and that the trajectory proceeds in order through the transitioning cluster 6, then through the early GC-like clusters 1 and 5 to the mature GC cluster 0 and on to cluster 2 or 9 (**Fig. 8H**). The analysis further indicated that early GC-like cluster 7 has a separate differentiation pathway, possibly indicating that loss of PTEN suppresses full differentiation of the GC-like tumor cells. These results are also consistent with our IHC marker analysis of tumor sections that showed that *SmoM2-Pten^{fl/fl}* tumors have an abundance of GC-like cells expressing NeuN and SYP, and further indicated such cells have in an early GC-like state (clusters 1, 5, 7).

We next performed differential expression analyses between *SmoM2* and *SmoM2-Pten^{fl/fl}* cells in the GCP-like tumor cell clusters (3, 4, 8) and found 1,011 genes were significantly upregulated and 3,710 were downregulated in the cells lacking *Pten* compared to *SmoM2* cells (adjusted p -value ≤ 0.05 , **Fig. 9A**, **Table S3**). Gene set enrichment analysis (GSEA) of Hallmark terms confirmed mTORC1 signaling is significantly unregulated in *SmoM2-Pten^{fl/fl}* tumors compared to

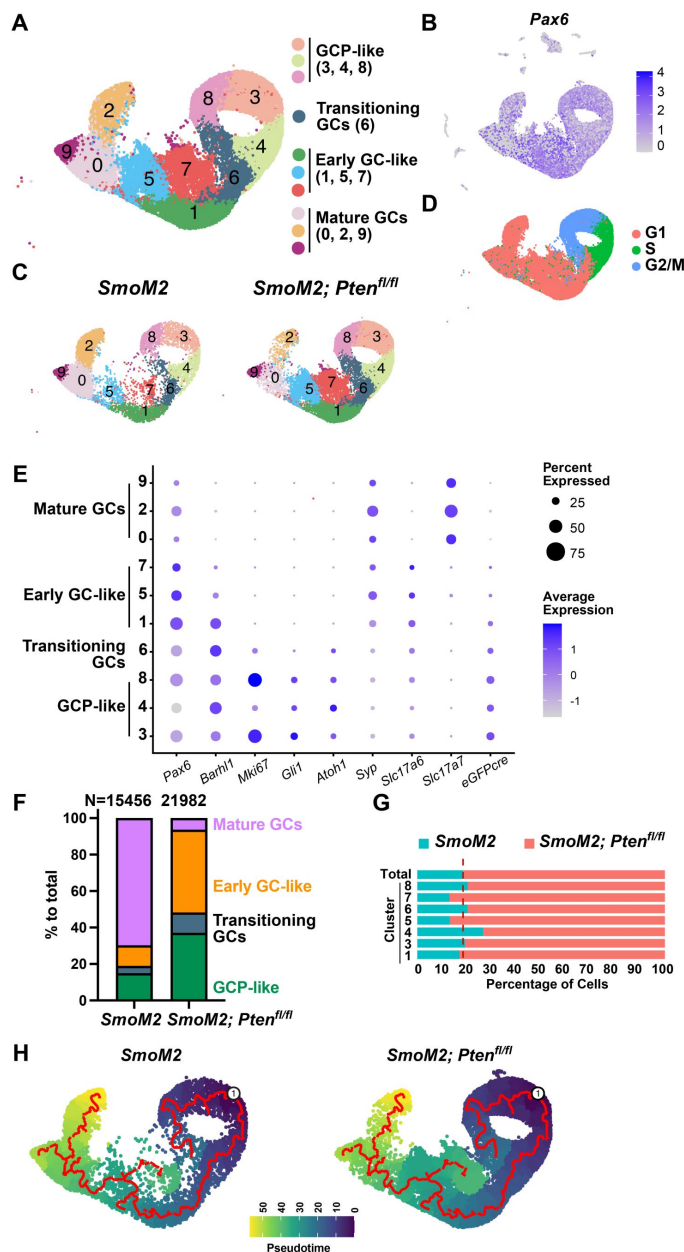


Figure 8.

scRNA-seq analysis of tumors from *SmoM2* and *SmoM2-Pten^{fl/fl}* mice reveals altered differentiation in SHH-MB tumor cells lacking *Pten*

(A) UMAP of integrated cells from *SmoM2* (n=2) and *SmoM2-Pten^{fl/fl}* (n=2) tumors showing scRNA-seq GCP-lineage clusters 0-9 (*Pax6*-expressing) (left) and the assignment of cell types to each cluster (right) based on marker gene expression and cell cycle phase. (B) *Pax6* expression (GCP-lineage marker) shown across all clusters in UMAP of all tumor cells. (C) UMAP of GCP-lineage cells shown by genotype. (D) UMAP of GCP-lineage cells (two genotypes) showing cell cycle phases. (E) Dot plot graph showing expression levels of GCP-lineage marker genes and designation of the clusters as GCP-like or maturing GC-like cell types. (F) Graph of GCP-lineage cells showing percentage of cells from each genotype that are GCP-like, transitioning GC-like, early GC-like and mature GC-like showing a higher percentage of mature GCs in tumors from *SmoM2* mice. (G) Graph showing proportions of cells of each genotype in each of the GCP-lineage cell types excluding mature GCs that are mostly from *SmoM2* mice. Dotted line represents the expected ratio between the two genotypes. (H) Pseudotime analysis reflecting the differentiation pathway of cells within the GCP-lineage with an origin in the GCP-like clusters, indicating cluster 7 is stalled or altered during differentiation.

SmoM2, as well as terms associated with MYC and E2F targets and unfolded protein response (Fig. 9B, Fig. 9-figure supplement 1A, Table S4). Violin plots of expression of mTORC1 genes supported upregulation of MTOR signaling (Fig. 9C). As for pathways downregulated in *SmoM2-Pten^{fl/fl}* cells compared to *SmoM2*, ‘Interferon alpha response’ was greatest, and the list included ‘complement’ and ‘inflammatory response’ (Fig. 9D), possibly contributing to the observed decrease in infiltrating macrophages in such tumors compared to *SmoM2*. Violin plots of genes involved in Interferon alpha signaling supported their down regulation in tumor cells lacking *Pten* (Fig. 9E). GSEA of Gene ontology (GO) terms (biological processes) showed upregulation of terms associated with neural development in *SmoM2-Pten^{fl/fl}* tumors compared to *SmoM2* (Fig. 9F, Fig. 9-figure supplement 1B, Table S5), indicating loss of PTEN pushes GCP-like cells toward neural differentiation. Violin plots of expression of genes involved in neural regulation supported upregulation of neurogenesis (Fig. 9G).

scRNA-seq analysis confirms reduced cell death

in differentiated cells of *SmoM2-Pten^{fl/fl}* tumors

Since our immunohistochemical analysis of *SmoM2-Pten^{fl/fl}* tumors showed that cell death was reduced in the NeuN+/SYP+ regions of tumors, we next performed differential expression analyses of only the *Pten* mutant cells in the early GC-like cluster 1 to the GCP-like clusters 3, 4 and 8 and found 2,738 genes were significantly upregulated and 4,185 were downregulated in cluster 1 cells compared to GCP-like cells (adjusted p-value<0.05, Fig. 9H, Table S6). GSEA of GO terms (biological processes) confirmed upregulation of terms associated with neural differentiation and also highlighted down regulation of ‘positive regulation of programmed cell death’ in *Pten* mutant cells in cluster 1 compared to the GCP-like clusters (Fig. 9I, K, Fig. 9-figure supplement 1C, Table S7). Violin plots of genes underlying the terms supported their altered expression in *SmoM2-Pten^{fl/fl}* tumors compared to *SmoM2* (Fig. 9J, L).

RNA in situ analysis of sections of P22 *SmoM2* and P16 *SmoM2-Pten^{fl/fl}* tumors was used to confirm some of the genes shown to be differentially expressed between the GCP-like cells in the two genotypes (Fig. 9M-R). *Asns* (Unfolded protein gene), *Vldlr* and *Cdkn1a* (mTORC1 pathway) showed upregulation in GCP-like tumor cells in P16 *SmoM2-Pten^{fl/fl}* compared to P22 *SmoM2* tumors.

scRNA-seq analysis indicates loss of *Pten* in *SmoM2*

SHH-MB tumor cells results in decreased expression

of genes associated with cytotoxicity in macrophages

Comparison of the transcriptomes of macrophages in *SmoM2-Pten^{fl/fl}* and *SmoM2* tumors indicated an anti-inflammatory phenotype in tumors lacking *Pten* because genes related to synaptic signaling and *Irf2* were decreased and *Cd36*, a marker of protumoral M2-type macrophages, was upregulated in *SmoM2-Pten^{fl/fl}* tumors (Fig. 9-figure supplement 1D, E, Table S8, S9).^{31,32} The observed reduction in macrophage infiltration into *SmoM2-Pten^{fl/fl}* tumors compared to *SmoM2* tumors and the apparent reduction in cytotoxicity might reflect an immunosuppressive tumor microenvironment that could contribute to the rapid growth of tumors lacking PTEN.

Discussion

Our work delineates cellular and transcriptional changes that lead to greatly accelerated tumor growth when *Pten* is mosaicly removed in GCPs that have activated SHH-signaling due to expression of *SmoM2*. We uncovered that in both wild type and *SmoM2* GCPs loss of *Pten* results in an initial increase in mitotic rate and likelihood to remain in the progenitor state. As *SmoM2*

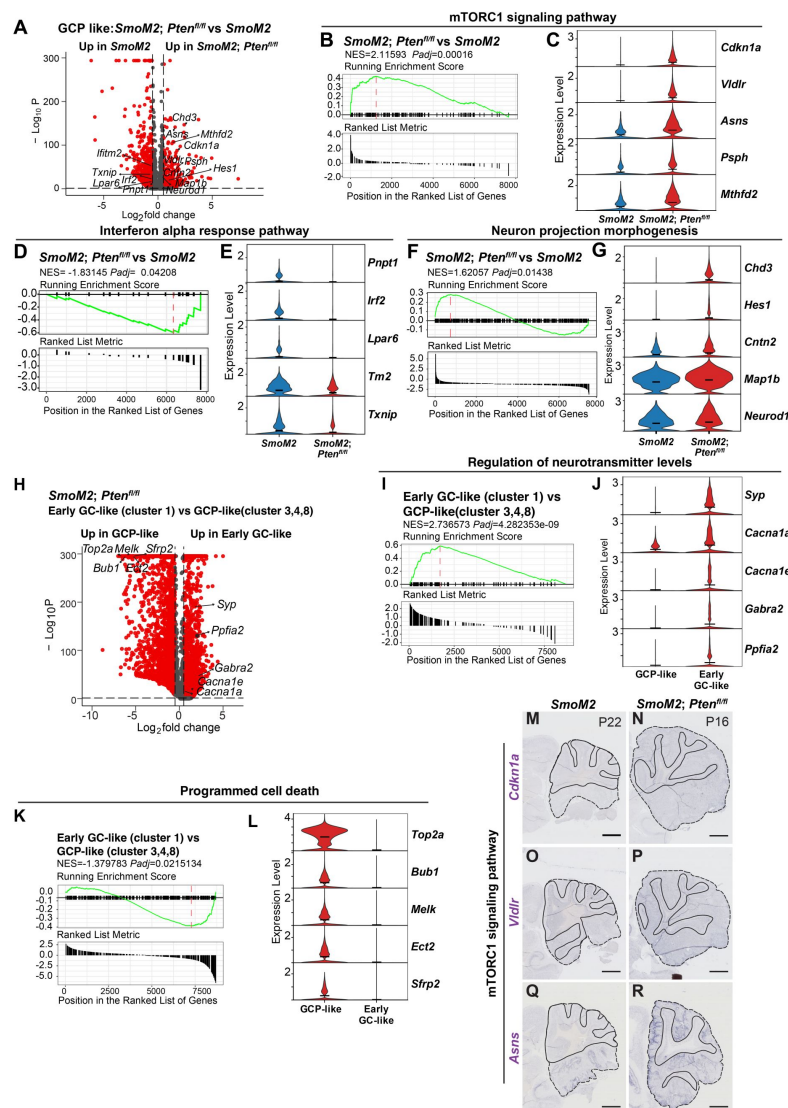


Figure 9.

SmoM2 tumors lacking *Pten* have altered neurogenesis and immune signaling

(A) Volcano plot showing differentially expressed genes in GCP-like tumor cell clusters (3, 4, 8) (red: adjusted p -value <0.05 , $ab[\log_2\text{fold-change}]>0.5$), in *SmoM2*-*Pten*^{fl/fl} tumors compared to *SmoM2*. (B) Gene set enrichment analysis (GSEA) plots (Hallmark terms) showing that mTORC1 signaling pathway is significantly upregulated in *SmoM2*-*Pten*^{fl/fl} tumors compared to *SmoM2*. (C) Violin plots showing expression levels of mTORC1 signaling pathway genes in the two genotypes. (D) GSEA plots (Hallmark terms) showing that Interferon alpha response is significantly downregulated in *SmoM2*-*Pten*^{fl/fl} tumors compared to *SmoM2*. (E) Violin plots showing expression levels of Interferon alpha response genes in the two genotypes. (F) GSEA plots of Gene ontology (GO) terms (biological processes) showing that regulation of neural differentiation is significantly upregulated in *SmoM2*-*Pten*^{fl/fl} tumors compared to *SmoM2*. (G) Violin plots showing expression levels of neural differentiation genes in the two genotypes, indicating an early stage of neurogenesis is enhanced. (H) Volcano plot showing differentially expressed genes in early GC-like cluster 1 compared to GCP-like cell clusters (3, 4, 8) in *SmoM2*-*Pten*^{fl/fl} tumors (red: adjusted p -value <0.05 , $ab[\log_2\text{fold-change}]>0.5$). (I) GSEA plots of Gene ontology (GO) terms (biological processes) confirmed upregulation of neural differentiation in cluster 1 compared to the GCP-like cell clusters. (J) Violin plots showing expression levels of neural differentiation genes in the two cell types. (K) GSEA plots of Gene ontology (GO) terms (biological processes) showed downregulation of 'positive regulation of programmed cell death' in cluster 1 compared to the GCP-like cell clusters. (L) Violin plots showing expression levels of programmed cell death genes in the two cell types. (M-R) Lateral sagittal sections of tumors from *SmoM2* mice (P22, M, O, Q) or *SmoM2*-*Pten*^{fl/fl} mice (P16, N, P, R) stained for RNA in situ analysis of *Cdkn1a*, *Vldlr*, *Asns*, indicating upregulation of mTORC1 signaling pathway in *Pten* mutant tumors.

tumor formation proceeds, tumors lacking *Pten* accumulate differentiated early GC-like cells that are stalled in their differentiation and are protected from cell death. Thus, the bulk of the *Pten* mutant tumors are post mitotic cells. Macrophages seem to avoid or do not readily penetrate the differentiated regions of the tumors, perhaps because there are less dying cells. With possible relevance to human tumors, unlike previous models of *Pten* heterozygous mutant mice that model Cowden Syndrome or hamartoma syndromes³³ or ones that involve embryonic conditional loss of *Pten* in all GCPs^{12, 13}, we found that heterozygous loss of *Pten* in sporadic GCPs does not result in more aggressive tumors than activating SHH-signaling alone in rare GCPs. Since our model more closely reflects the sporadic nature of human tumor mutations, our results predict that heterozygous loss of *PTEN* will primarily have a negative impact on disease outcome when the mutations are germline (*PTEN*^{+/−}). If human *PTEN* mutant SHH-MB have a similar differentiated cellular phenotype to mouse models, then targeting such tumors with drugs to kill proliferating cells will not remove the bulk of the tumor and will require the drugs having access to the rare proliferating GCP-like tumor stem-like cells.

We found that loss or reduction of *Pten* in *SmoM2* mice initially increases expansion of proliferating GCPs through increased proliferation and decreased differentiation such that by P8 (one week after gene mutation at P0) the EGL is expanded in *SmoM2-Pten*^{fl/+} and *SmoM2-Pten*^{fl/fl} mice compared to *SmoM2*. By P12, however the EGL expansion is greatly enhanced only in *SmoM2-Pten*^{fl/fl} mice, and this is accompanied by an abundance of differentiated cells and a significant reduction in cell death. Our label retention assay involving BrdU injection at P10 revealed that a greater proportion of BrdU⁺ *SmoM2-Pten*^{fl/fl} GCPs were Ki67[−] at P12 than in *SmoM2* mice. This result indicates either that by P10 the loss of *Pten* results in an increase in differentiation of GCPs compared to in *SmoM2* mice, which would be consistent with our scRNA-seq analysis, but also likely due to a greater survival of the differentiated cells since the density of TUNEL⁺ particles is reduced in the EGL of *SmoM2-Pten*^{fl/fl} mice at P12. Consistent with this, it was shown that NeuN⁺ cells in SHH-MB tumors preferentially die compared to proliferating or quiescent tumor cells in a *Ptch1* mutant model²⁶. This previous result is the opposite to what we observed in late stage *SmoM2-Pten*^{fl/fl} tumors where the proliferative compartments of the tumors has greater cell death than the NeuN⁺ or SYP⁺ regions, and a similar density of TUNEL particles to in *SmoM2* tumors. Thus, a key reason for the accelerated tumor growth in *Pten* mutant SHH-MB models is the survival of the differentiated compartment.

In a mouse *NeuroD2-SmoA1* model of SHH-MB *Pten* heterozygosity leads to earlier onset of disease and a differentiated/nodular cellular phenotype when *Pten* is deleted in the germline or conditionally throughout the brain (*Nestin-Cre*)¹². Heterozygous loss of *Pten* in GCPs using an *Atoh1-Cre* transgene to delete *Pten*^{fl/+} and *Ptch1*^{fl/fl} apparently also slightly increases the time of onset and produces a differentiated phenotype¹³. In contrast, in our sporadic SHH-MB model (*SmoM2-Pten*^{fl/+}) where only rare GCPs lack one copy of *Pten* (and express *SmoM2*), the time of onset and penetrance are not accelerated and the tumors have a classic histology. Only in rare *SmoM2-Pten*^{fl/+} clones of cells in a minority of tumors where PTEN protein is lacking, likely due to loss of heterozygosity, did we observe a differentiated phenotype with reduced cell death as seen in *SmoM2-Pten*^{fl/fl} mice. The difference in the outcome of our model and the other two is either because of the SHH-activating mutations (*SmoM2* vs *SmoA1* expression or *Ptch1* loss) or more likely because when all GCPs lack a copy of *Pten* and have activated SHH-signaling that the microenvironment provides an advantage for *Pten* heterozygous cells to form a differentiated tumor. It is also possible that another cell type targeted by the other two *Cre* transgenes (*Netstn* and *Atoh1*) supports tumor growth when *Pten* loss is heterozygous.

A previous study provided evidence that increased mTOR signaling in SHH-MB promotes spinal cord metastasis by expressing *Shh* in GCPs along with an activated form of AKT^{18, 19}. However, in our sporadic model and the two other models of *Pten* loss in mouse SHH-MB^{12, 13}, spinal cord metastasis is not increased. Thus it appears that the way in which the mTOR pathway is activated affects metastasis. Perhaps related to this, in a model of *Trp53* heterozygous loss and

Rictor homozygous loss, which reduces pAKT, GCPs were found to persist longer than normal in the EGL³⁴, which is the same as what we found when *Pten* was deleted. Thus, the way the pathway downstream of PTEN is altered dictates the subsequent behavior of the mutant GCPs.

In our previous study we showed that in *Kmt2d* mutant SHH-MB models, no difference in tumor size was detected at P12 but by P21 tumors lacking one or two copies of *Kmt2d* were more than twice the size of those with intact *Kmt2d* and had a classic cytoarchitecture. Thus, there is a delayed positive effect of reducing/ablating this chromatin modifier gene on tumor growth rate compared to loss of *Pten* where there is a rapid expansion of the EGL. Another major difference between secondary loss of the two tumor suppressors in SHH-MB is that *Kmt2d* mutant tumors have nearly fully penetrant spinal cord metastasis. Thus, continued mechanistic studies in mouse models of recurrent mutations seen in SHH-MB should provide valuable insights into how the disease is expected to progress and provide ideas for development of targeted therapies based on the cellular phenotypes of tumors not just the signaling pathways altered.

Materials and methods

Animals

The animal care and procedures were performed in accordance with the Memorial Sloan Kettering Cancer Center Institutional Animal Care and Use Committee guidelines (IACUC protocol: 07-01-001). Mice were kept on a 12/12 hour light/dark cycle and in temperature- and humidity-controlled rooms with *ad libitum* access to standard laboratory mouse chow and water. All transgenic mouse lines were maintained on a mixed genetic background containing 129, C57BL/6J, and Swiss Webster. Both sexes were used for the analysis. The following mouse lines were used in the study: *Atoh1-Cre* (JAX #011104), *R26^{LSL-eGFPcre}* (MASTR, JAX #018903)²², *Atoh1-FlpoER* (JAX #040091)²³, *Pten^{fl/fl}*²⁵ and *SmoM2* (#005130)²⁴. Genetic recombination was induced with FlpoER by injecting one 100µg/g dose of Tamoxifen (Tm) (Sigma-Aldrich) subcutaneously into the back of P0 mice. Tm was prepared in corn oil (Sigma-Aldrich) at 20 mg/mL and stored at 4°C.

Tissue preparation

For all histological analyses, mice were anesthetized with isoflurane and then perfused transcardially with room temperature (RT) phosphate buffered saline (PBS) with heparin (0.02 mg/mL) and then ice cold 4% paraformaldehyde (PFA, Electron Microscopy Sciences, catalog no: 15714). Brains and spinal cords were dissected and post-fixed in 4% PFA 3–4 days at 4°C and cryopreserved in 30% sucrose in PBS for ~2 days at 4°C. Brains were embedded in Tissue-Tek OCT compound (Sakura Finetek). Brains were serially cryosectioned sagittally at 12 µm and collected on charged glass slides (Fisherbrand ColorFrost Plus) and stored at -20°C. Frozen spinal cords were coronally sectioned at 20µm. Details of reagents are listed in **Key Resources Table**.

Hematoxylin and eosin (H&E) staining

H&E staining were obtained from Richard-Allan Scientific and included hematoxylin 2 solution, eosin-Y, bluing reagent, and clarifier 2. Slides were rinsed in PBS for 5 min and incubated in hematoxylin 2 solution for 3 min and then rinsed in deionized water (diH₂O) and placed in staining reagents for 1 min each (diH₂O, clarifier 2, diH₂O, bluing reagent, diH₂O, eosin-Y). Dehydration and defatting were then performed (diH₂O–70% ethanol–95%–95%–100%–100%–xylene–xylene–xylene, 1 min each) and then slides were mounted with a coverslip and DPX mounting medium (Electron Microscopy Sciences). Details of reagents are listed in **Key Resources Table**.

Immunofluorescence

Slides were washed 3 times for 5 min in PBS. When necessary, antigen retrieval was then performed by immersing slides in sodium citrate buffer (10 mM sodium citrate with 0.05% Tween-20, pH 6.0) for 20 min at 95°C followed by rinsing in PBS. Slides were then incubated in blocking buffer (5% BSA in 0.4% Triton-X100 in PBS (PBST)) for 1 hour at RT. Slides were then placed in primary antibody solution in blocking buffer at 4°C overnight. Primary antibody information and dilutions are listed in **Key Resources Tables**. Slides were then rinsed three times in RT 0.1% PBST and incubated in secondary antibody solution in blocking buffer (1:500) at RT for 1 hour. Slides were then incubated in Hoechst (Invitrogen, catalog #H3569 diluted 1:1000 in PBS) for 10 min and rinsed three times in PBS and cover-slipped with Fluoro-Gel (Electron Microscopy Sciences).

TUNEL Staining

TUNEL staining was according to the manufacturers instructions (Roche TUNEL assay kit). Tissue on slides were permeabilized with 0.5% TritonX-100 and then pre-incubated with 1x TdT reaction buffer with 5mM CoCl₂ and 1X PBS for 15 min at RT. Slides were then incubated for 1 hr at 37°C in 1x TdT reaction buffer containing Terminal Transferase (Roche) and Biotin-16-dUTP (Sigma-Aldrich) or Digoxigenin-dUTP (Enzo Life Sciences, Inc). Slides were then incubated with a Streptavidin Alexa Fluor 647 conjugate (Invitrogen catalog #S-32357) or Anti-Digoxigenin-Rhodamine (Sigma-Aldrich catalog), respectively for 1 hr.

BrdU Injection and Staining

To assess cell differentiation, BrdU (Sigma Aldrich) was injected intraperitoneally at 50 µg/g at P10 for analysis at P12. Tissues were collected and processed as described above. For immunostaining, antigen retrieval was applied to slides with Rat primary anti-BrdU antibody BU1/75 (ICR1) (Abcam Inc.) to detect BrdU.

EdU (5-ethynyl-2'-deoxyuridine) Injection and Staining

To assess cell proliferation, EdU (Invitrogen) was injected intraperitoneally at 50 µg/g 1 hr before euthanasia. A Click-it EdU assay was used (Invitrogen) with Sulfo-Cyanine5 azide (Lumiprobe Corporation) per the manufacturer's protocol to stain sections.

RNA *in situ* hybridization

Sagittal sections (12 µm) were processed for *in situ* hybridization as described³⁵. Except for *Gli1* and *Syp*, antisense riboprobes were labeled with UTP-digoxigenin using a DIG RNA labeling mix (Sigma-Aldrich) with PCR-amplified templates prepared from cDNA synthesized from an early postnatal mouse cerebellum lysate. Signal was detected with alkaline phosphatase-coupled anti-digoxigenin antibodies using BM purple (Sigma-Aldrich) as the substrate. The primers used for PCR amplification were based on the Allen Brain Atlas. *Gli1* RNA *in situ* was performed using an antisense RNA probe made from a plasmid³¹ and for *Syp* BioMatik generated a plasmid in Bluescript containing *Syp* sequences using the primers described in the **Key Resources Tables**. Primers were flanked in the 3' with SP6 (antisense) and 5' with T3 (sense) promoters. Primers for RNA *in situ* probes are listed in **Key Resources Tables**.

Microscopy, image processing, and analyses

Images were acquired with a DM6000 Leica fluorescent microscope (Leica Camera, Wetzlar, Germany) or NanoZoomer Digital Pathology microscope (Hamamatsu Photonics, Shizuoka, Japan) and were processed and analyzed using Fiji³⁶ or Photoshop (Adobe Inc., San Jose, CA, USA). Cell counts were manually obtained using the Cell Counter plugin for Fiji or semi-automated using a custom script using the Analyze Particle plugin for Fiji as previously described³⁷.

ScRNA-seq sample Preparation

Atoh1-FlpoER/+; R26^{MASTR/LSL-SmoM2};Pten^{+/-} (SmoM2) and *Atoh1-FlpoER/+; R26^{MASTR/LSL-SmoM2}; Pten^{fl/fl}* (SmoM2 *Pten^{fl/fl}*) mice were euthanized at P16 and P22, respectively, and the tumors were dissected from the lateral-posterior cerebellum in ice-cold Hank's balanced salt solution (HBSS, Gibco). Then the tumors were dissociated in Papain (Worthington) for 30 minutes at 37°C before being washed out in Ovalbumin (Worthington). Tumors were triturated by pipetting up and down using a P1000 and filtered through a 40µm cell strainer (Falcon). Once the cells were dissociated, the dead cells were removed using the kit EasySep Dead Cell Removal (Annexin V) (StemCell). Then cells were centrifuged at 500g at 4°C for 5 minutes. Cells were then washed with PBS. For every sample (2 *SmoM2* and 2 *SmoM2 Pten^{fl/fl}*), cells were counted using 0.2% Trypan on hemocytometer and diluted before loading to 10X chip, targeting 10,000 viable cells per sample. The scRNA-Seq of tumor cell suspensions was performed on a Chromium instrument (10X genomics) following the user guide manual for 3' v3.1. Final libraries were sequenced on an Illumina NovaSeq6000 by the IGO core of MSKCC.

scRNA-seq data analysis

The Cell Ranger Single Cell software suite (10x Genomics) was used to align reads and generate feature-barcode matrices. The reference genome used was the Genome Reference Consortium Mouse Build 38 (GRCm38, Gencode annotation mm10). Raw reads were processed using the Cell Ranger count program using default parameters.

Seurat v5.0.2 package was used to generate a UMI (unique molecular identifier) count matrix from the Cell Ranger output³⁸. Genes expressed in less than 10 cells were removed for further analyses. Cells with larger than 18000 UMIs, 0.20 mitoRatio, UMI≤500 and ≤500 genes detected were considered low quality/outliers and discarded from the datasets. Normalization was performed on individual samples using the NormalizeData function with default parameters. The normalized data was scaled by ScaleData function with mitochondrial gene percentage regressed out. Principal component analysis (PCA) was performed on the scaled data by runPCA. Samples were then integrated using IntegrateLayers function with the Harmony Integration method. The first 20 dimensions were used for the FindNeighbors function, and clusters were identified using the FindClusters function with a resolution of 0.5. Data were projected onto the 2D space using the FindUMAP or RunUMAP function with 20 dimensions. Cluster markers and further differential gene expression analyses were performed using normalized counts (NormalizeData) in the RNA assay. Cluster markers were identified using the FindAllMarkers and comparing markers generated to the literature. To refine clustering further, the SubsetData function was used to create a new Seurat object, and the above clustering was reiterated.

Differential gene expression analyses between genotypes or clusters were performed using the FindMarkers function in the Seurat package. Genes with an adjusted *P* value (*Padj*) < 0.05 were considered significantly up or downregulated. Results were visualized by Violin plot and Umap plot using Seurat package, and Volcano plot using the EnhancedVolcano package on Bioconductor³⁹. GSEA analysis on GO and Hallmark terms were performed using the enrichGO, clusterProfiler packages on Bioconductor⁴⁰. To build a pseudotime trajectory using the UMAP data, the Seurat object was converted to a monocle 3 (1.3.7) object by as.cell_data_set function²⁹. Cells were then clustered by the cluster cells function. The trajectory GCP-lineage partition was constructed using learn graph function followed by order cells function.

Statstic analysis

All statistical analyses to quantify data were performed using Prism software (GraphPad) and significance was determined as *P* ≤ 0.05. Each statistical analysis used is described in the figure legends. Data (shown as scatter graphs) were presented as mean±standard deviation.

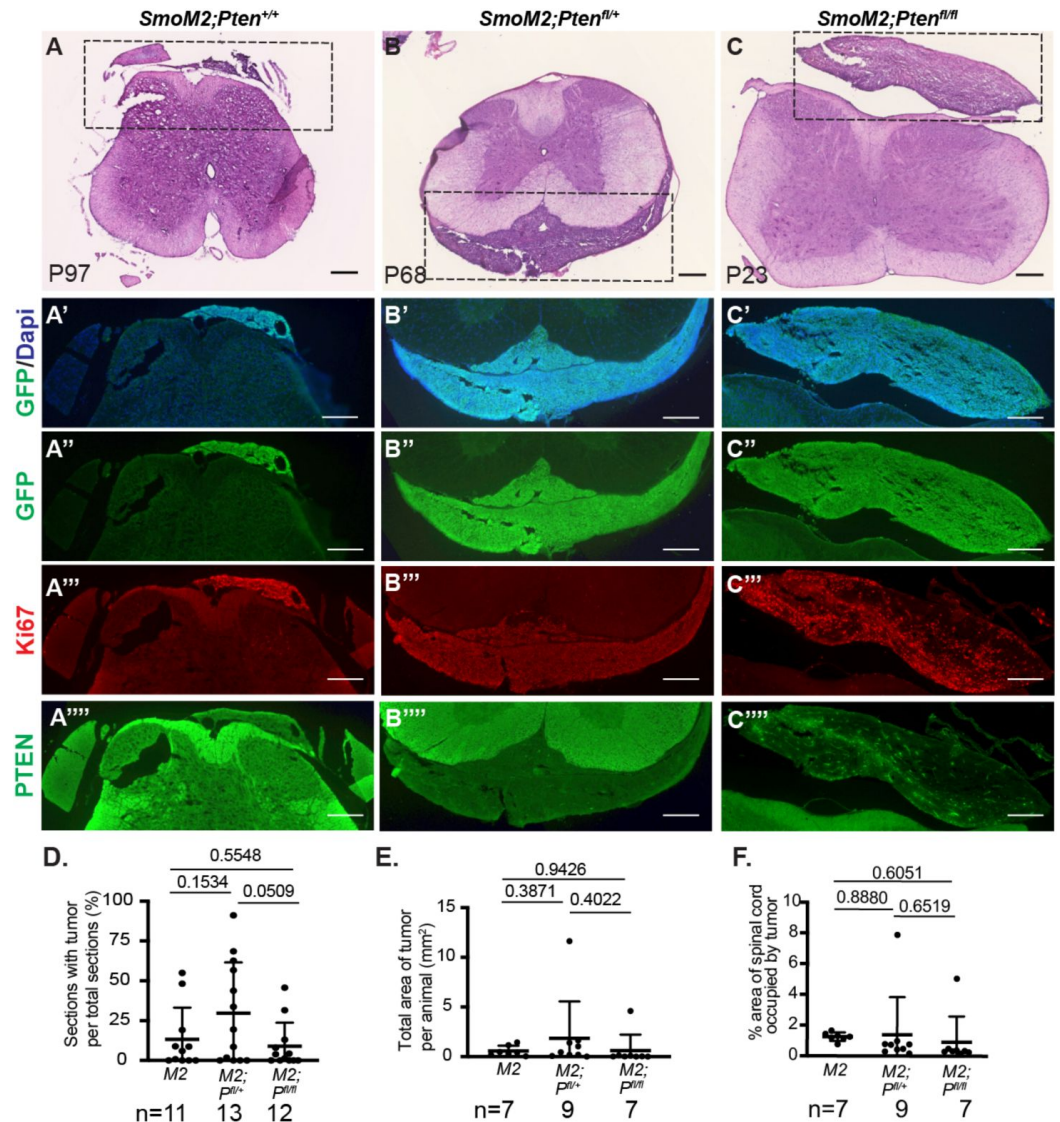


Figure 1-Figure Supplement 1.

Loss of *Pten* does not increase metastasis to the spinal cord in *SmoM2* SHH-MB

(A-C) Coronal sections of spinal cords stained with H&E from mice of the indicated three genotypes showing metastatic lesions at end stage. Dashed rectangles indicate locations of images below stained for the proteins and DAPI indicated on the left. Scale bars indicates 200µm. **(D)** Quantification of percentage of sections with a tumor for each animal analyzed. **(E)** Quantification of the total area of the tumor on all the sections. **(F)** Quantification of the area of each tumor as a percentage of the spinal cord area in each section. (D-F) Number of animals per genotype (n=) are indicated below graphs. All statistics were determined using an unpaired t-test comparing *SmoM2* tumors to each of the *Pten* mutants or the *Pten* mutants to each other.

Figure 2–Figure Supplement 1.

Spontaneous loss of the second allele of *Pten* in rare cells of some *SmoM2-Pten^{fl/+}* tumors

(A–D) Sagittal sections of end stage tumors (GFP+) showing upregulation of pS6 in *SmoM2-Pten^{fl/fl}* mice compared to *SmoM2* (n=3 per genotype). Dashed lines outline the tumors, solid lines outline IGL. Scale bars indicates 1mm. (E) Section of end stage tumor from *SmoM2-Pten^{fl/+}* mice showing coherent patches of tumor cells (GFP+) devoid of PTEN (n=3). Dashed line square indicates location of high magnification images shown in (F–L). Scale bar indicates 1mm. (F–L) Sections of lateral tumors stained for the indicated protein and DAPI (n=3). Dashed line outlines tumor area where PTEN is absent (G) or pAKT (H), SYP (J) and NeuN (K) are upregulated and proliferation (Ki67) and cell death (TUNEL) are reduced (I, L). Scale bars indicates 200µm.

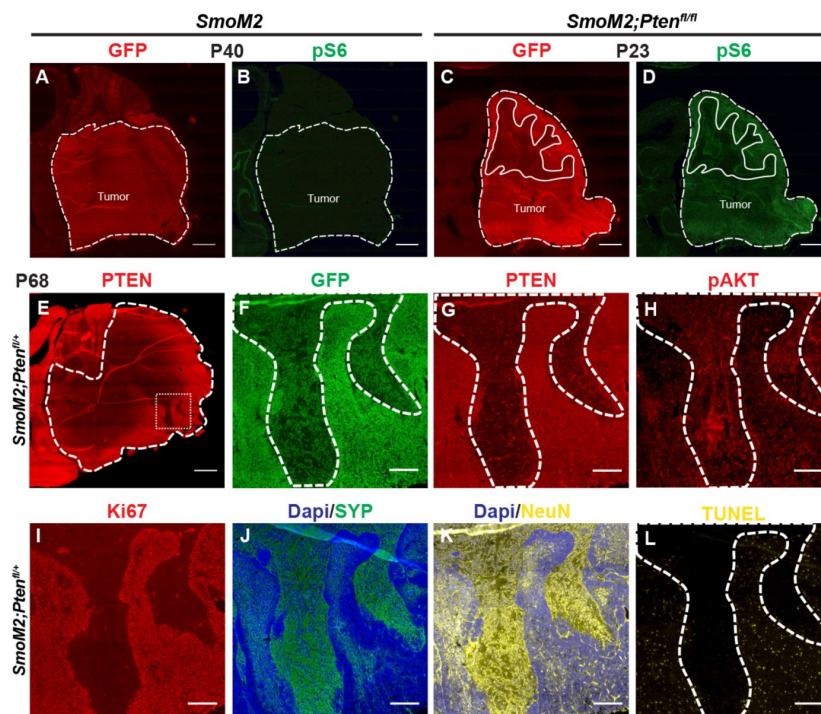
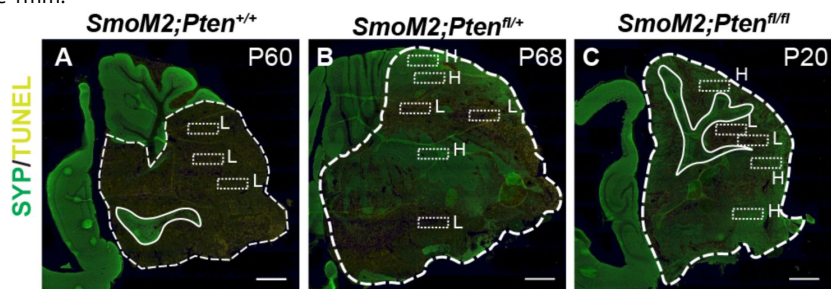


Figure 3–figure supplement 1.

Method of quantifying TUNEL particles in tumors

(A–C) Examples of stained sagittal cerebellar sections showing where quantifications of density of TUNEL particles were performed in the three genotypes in end stage lateral tumor regions distinguishing high SYP (H) compared to low SYP (L) regions. Dashed lines outline each tumor, solid lines outline IGL, dotted boxes indicate high SYP (H) and low SYP (L) regions. Scale bars indicate 1mm.



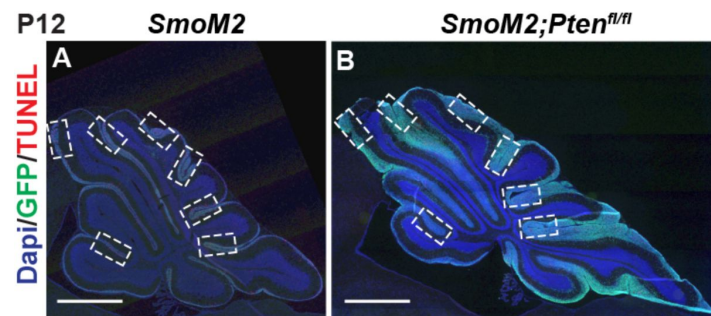


Figure 4-figure supplement 1.

Method of quantifying TUNEL particles in P12 EGL

(A, B) Examples of stained midline sagittal cerebellar sections showing where quantification of density of TUNEL particles was performed in the EGL in medial regions of the cerebellum. Scale bars indicate 1mm.

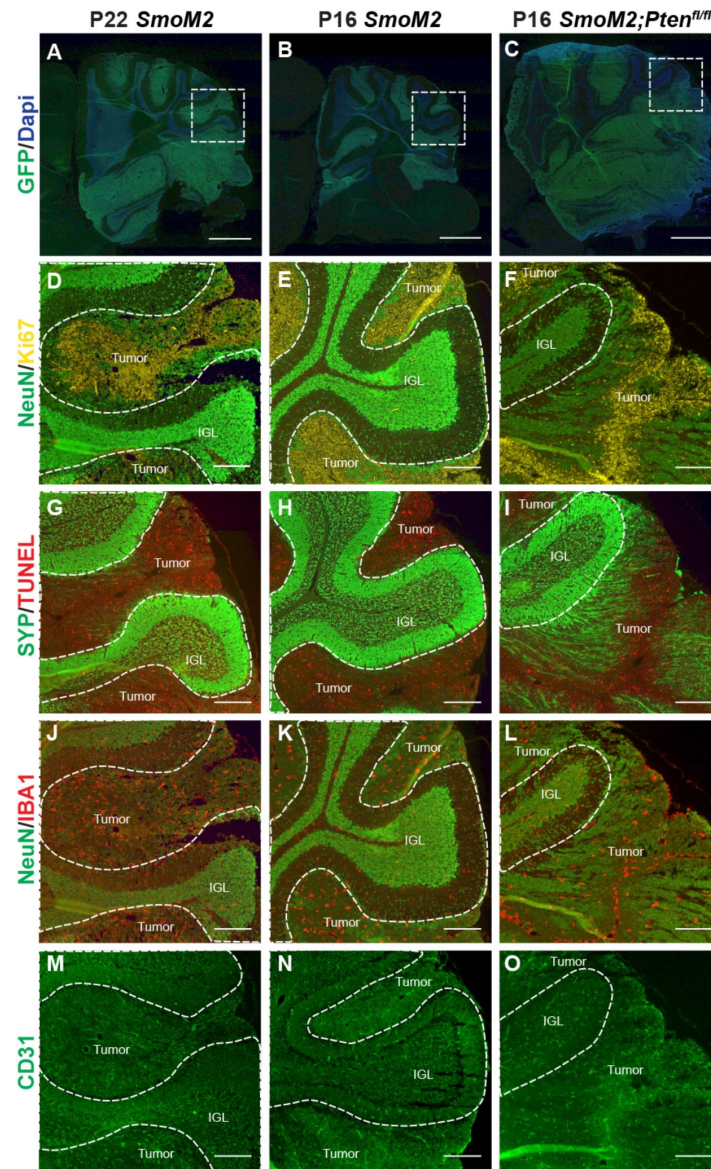


Figure 8-figure supplement 1

The cellular phenotypes of large tumors from P16 *SmoM2-Pten^{fl/fl}* mice and P22 *SmoM2* mice are similar to end stage

(A-C) Sagittal sections of cerebellar sections from P16 and P22 *SmoM2* mice and P16 *SmoM2-Pten^{fl/fl}* mice stained with GFP and DAPI to highlight tumor cells (n=3 mice/genotype). Dashed line squares indicate location of high magnification images shown in (D-O). Scale bars indicate 1mm. (D-O) Higher magnification images of tumors stained for indicated proteins to confirm that the cellular phenotypes of large tumors in P16 *SmoM2-Pten^{fl/fl}* mice and P22 *SmoM2* mice are like tumors at end stage. Dashed lines outline the internal granule layer (IGL). Scale bars indicate 200µm.

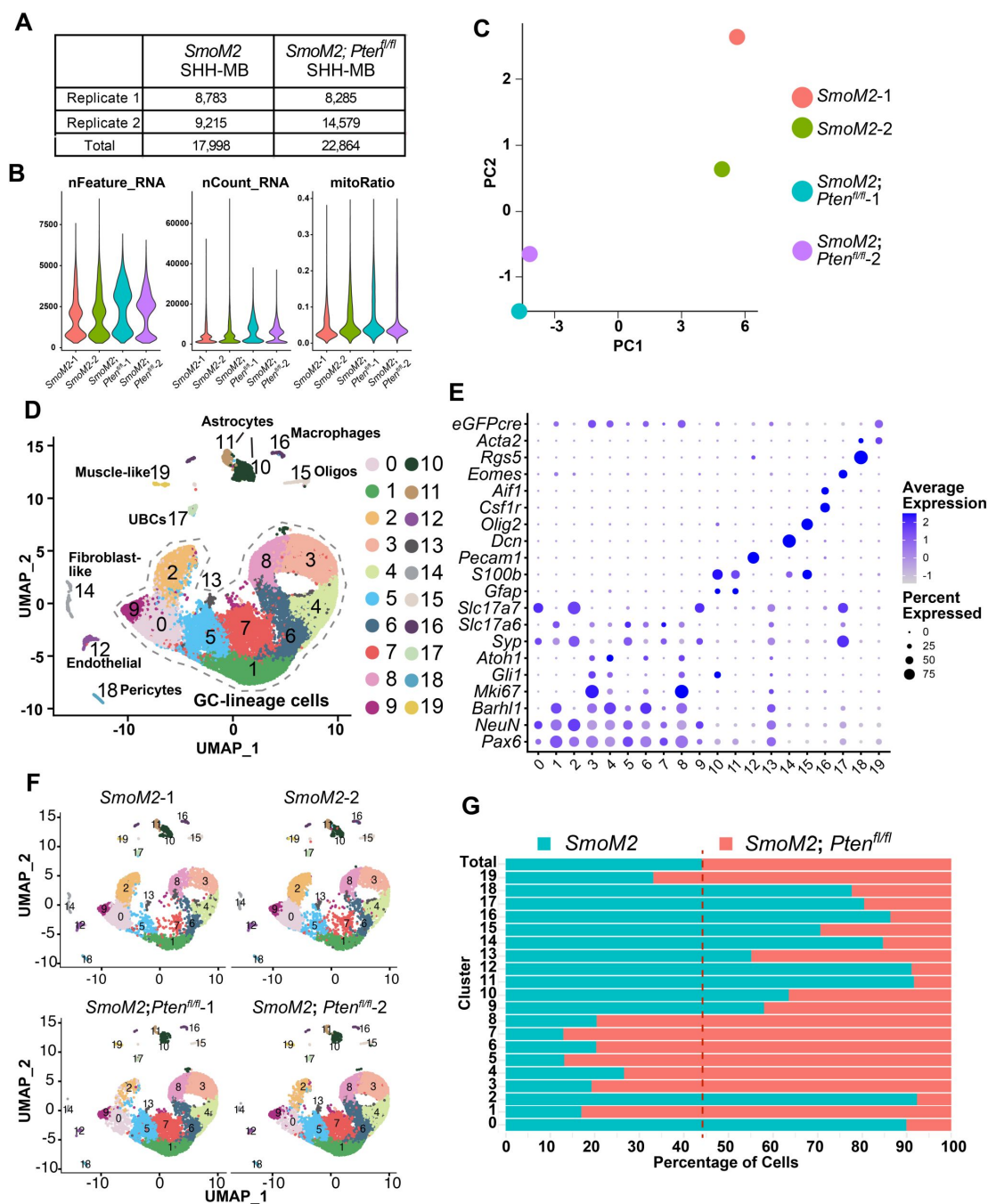


Figure 8-figure supplement 2.

ScRNA-seq comparison of tumors from P16 *SmoM2-Pten^{fl/fl}* mice to P22 *SmoM2* mice

(A) Number of cells from each replicate and genotype used for downstream analyses after filtering. (B) Violin plots showing the number of features, and RNA counts and percent mitochondrial RNA counts across the biological replicates of the scRNA-seq data set after filtering out poor quality cells (where number of detected genes was ≤ 500 , the number of detected transcripts was $\geq 18,000$ and mitochondrial gene percentage $\geq 20\%$). (C) PCA plot after filtering out poor quality cells and integrating the replicates and genotypes showing the two genotypes separate well. (D) UMAP showing projections of all cells showing cluster annotations. (E) Dot plot graph showing expression levels of cell type marker genes across all clusters and their cell type designations. (F) UMAPs of all cells shown by genotype and sample. (G) Graph showing proportions of cells of each genotype when all cells are included. Dotted line represents the expected ratio between the two genotypes.

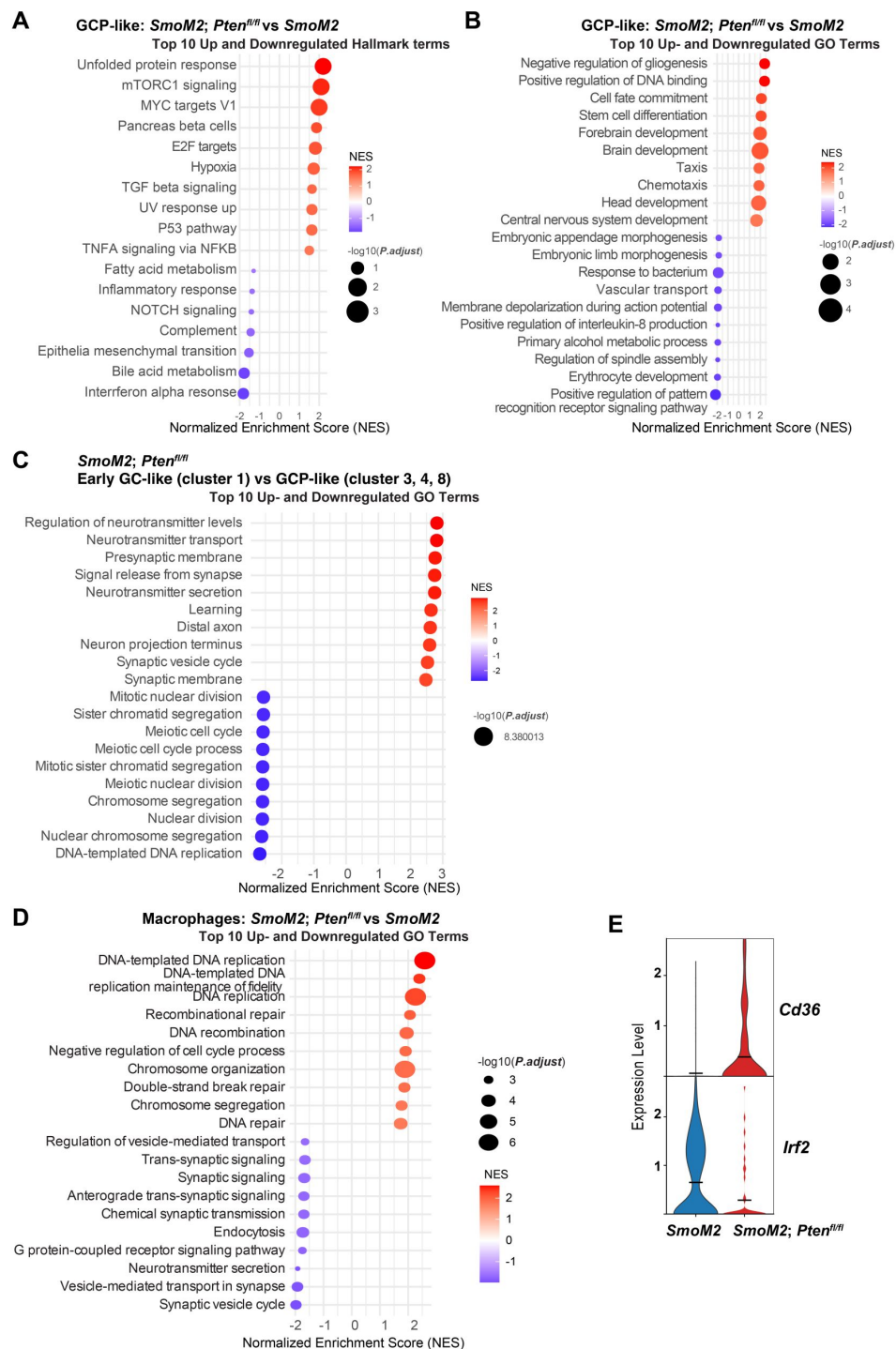


Figure 9-figure supplement 1.

Differential expression analyses confirms activation of mTOR

A. Bar plots showing top 10 upregulated and downregulated Hallmark terms in *SmoM2*-*Pten*^{fl/fl} GCP-like cells compared to *SmoM2*. **B.** Bar plots showing top 10 upregulated and downregulated Gene ontology (GO) terms (biological processes) in GCP-like cells from *SmoM2*-*Pten*^{fl/fl} mice compared to *SmoM2*. **C.** Bar plots showing top 10 upregulated and downregulated Gene ontology (GO) terms (biological processes) in early GC-like cluster 1 compared to GCP-like cell clusters (3, 4, 8) in *SmoM2*-*Pten*^{fl/fl} tumors. **D.** Bar plots showing the top 10 upregulated and downregulated Gene ontology (GO) terms (biological processes) in macrophages in *SmoM2*-*Pten*^{fl/fl} tumors compared to *SmoM2*. **E.** Violin plots showing expression levels of *Cd36* and *Irf2* in macrophages in the two tumor genotypes.

Data Availability

Sequencing data have been deposited in GEO (token czkvuymyhnktrgp) under accession code GSE295733. All experimental data generated and analyzed in this study are included in the manuscript and supporting files.

Acknowledgements

We thank members of the Joyner lab for interesting discussions pertaining to SHH-MB and technical support, in particular Reeti Mayur Sanghrajka for insightful suggestions at the beginning of the study. We thank the Memorial Sloan Kettering Cancer Center Flow Cytometry Core, Integrated Genomics Operations Core Facility and the Center for Comparative Medicine and Pathology for technical support.

Additional information

Funding

NCI (R01CA192176), Alexandra L Joyner

MSK Functional Genomics Initiative (GC242211), Alexandra L Joyner

NCI Cancer Center Support Grant (CCSG, P30 CA08748), Alexandra L Joyner Cycle for Survival, Alexandra L Joyner

Ethics

No ethical consideration as no human data were analyzed.

Author contributions

Z.L., S.E.N., Y.L and A.L.J. formulated experiments and analysis approaches. Z.L. and D.N.S. performed animal and histological experiments. Z.L. performed section analysis and quantifications. Z.L., S.E.N. and Y.L. performed scRNA-seq analysis. A.L.J. wrote the draft manuscript and Z.L., Y. L., S.E.N. and A.L.J. prepared Figures and edited the manuscript.

Additional files

Key Resources Table [↗](#)

Supplementary Tables S1-9 [↗](#)

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Author information

Zhimin Lao

Developmental Biology Program, Sloan Kettering Institute, Memorial Sloan Kettering Cancer Center, New York, United States

Salsabiel El Nagar[#]

Developmental Biology Program, Sloan Kettering Institute, Memorial Sloan Kettering Cancer Center, New York, United States, Sophrosyne Bio, New York, United States

[#]These authors contributed equally to the work

Yinwen Liang[#]

Developmental Biology Program, Sloan Kettering Institute, Memorial Sloan Kettering Cancer Center, New York, United States

[#]These authors contributed equally to the work

Daniel N Stephen

Developmental Biology Program, Sloan Kettering Institute, Memorial Sloan Kettering Cancer Center, New York, United States

Alexandra L Joyner

Developmental Biology Program, Sloan Kettering Institute, Memorial Sloan Kettering Cancer Center, New York, United States, Biochemistry, Cell and Molecular Biology Program, Weill Cornell Graduate School of Medical Sciences, New York, United States

ORCID iD: [0000-0001-7090-9605](https://orcid.org/0000-0001-7090-9605)

For correspondence: joynera@mskcc.org

Editors

Reviewing Editor

Lynne-Marie Postovit

Queens University, Kingston, Canada

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Lynne-Marie Postovit

Queens University, Kingston, Canada

Reviewer #1 (Public review):

Summary:

This paper investigates how Pten loss influences the development of medulloblastoma using mouse models of Shh-driven MB. Previous studies have shown that Pten heterozygosity can accelerate tumorigenesis in models where the entire GNP compartment has MB-promoting mutations, raising questions about how Pten levels and context interact, especially when cancer-causing mutations are more sporadic. Here, the authors create an allelic series combining sporadic, cell-autonomous induction of SmoM2 with Pten loss in granule neuron progenitors. In their models, Pten heterozygosity does not significantly impact tumor development, whereas complete Pten loss accelerates tumour onset. Notably, Pten-deficient tumours accumulate differentiated cells, reduced cell death, and decreased macrophage

infiltration. At early stages, before tumour establishment, they observe EGL hyperplasia and more pre-tumour cells in S phase, leading them to suggest that Pten loss initially drives proliferation but later shifts towards differentiation and accumulation of death-resistant, postmitotic cells. Overall, this is a well-executed and technically elegant study that confirms and extends earlier findings with more refined models. The phenotyping is strong, but the mechanistic insight is limited, especially with respect to dosage effects and macrophage biology.

Strengths:

The work is carefully executed, and the models-using sporadic oncogene induction rather than EGL-wide genetic manipulations-represent an advance in experimental design. The deeper phenotyping, including single-cell RNA-seq and target validation, adds rigor.

Weaknesses:

The biological conclusions largely confirm findings from previous studies (Castellino et al, 2010; Metcalf et al, 2013), showing that germline or conditional Pten heterozygosity accelerates tumorigenesis, generates tumors with a very similar phenotype, including abundant postmitotic cells, and reduced cell death.

The second stated goal - to understand why Pten dosage might matter - remains underdeveloped. The difference between earlier models using EGL-wide SmoA1 or Ptch loss versus sporadic cell-autonomous SmoM2 induction and Pten loss in this study could reflect model-specific effects or non-cell-autonomous contributions from Pten-deficient neighbouring cells in the EGL, for example. However, the study does not explore these possibilities. For instance, examining germline Pten loss in the sporadic SmoM2 context could have provided insight into whether dosage effects are cell-autonomous or dependent on the context.

The observations on macrophages are intriguing but preliminary. The reduction in Iba1+ cells could reflect changes in microglia, barrier-associated macrophages, or infiltrating peripheral macrophages, but these populations are not distinguished. Moreover, the functional relevance of these immune changes for tumor initiation or progression remains unexplored.

<https://doi.org/10.7554/eLife.108190.1.sa1>

Reviewer #2 (Public review):

The authors sought to answer several questions about the role of the tumor suppressor PTEN in SHH-medulloblastoma formation. Namely, whether Pten loss increases metastasis, understanding why Pten loss accelerates tumor growth, and the effect of single-copy vs double-copy loss on tumorigenesis. Using an elegant mouse model, the authors found that Pten mutations do not increase metastasis in a SmoD2-driven SHH-medulloblastoma mouse model, based on extensive characterization of the presence of spinal cord metastases. Upon examining the cellular phenotype of Pten-null tumors in the cerebellum, the authors made the interesting and puzzling observation that Pten loss increased the differentiation state of the tumor, with fewer cycling cells, seemingly in contrast to the higher penetrance and decreased latency of tumor growth.

The authors then examined the rate of cell death in the tumor. Interestingly, Pten-null tumors had fewer dying cells, as assessed by TUNEL. In addition, the tumors expressed differentiation markers NeuN and SyP, which are rare in SHH-MB mouse models. This reduction in dying cells is also evident at earlier stages of tumor growth. By looking shortly after Pten-loss induction, the authors found that Pten loss had an immediate impact on increasing the proliferative state of GCPs, followed by enhancing the survival of

differentiated cells. These two pro-tumor features together account for the increased penetrance and decreased latency of the model. While heterozygous loss of Pten also promoted proliferation, it did not protect against cell death.

Interestingly, loss of Pten alone in GCPs caused an increase in cerebellar size throughout development. The authors suggest that Pten normally constrains GCP proliferation, although they did not check whether reduced cell death is also contributing to cerebellum size.

Lastly, the authors examined macrophage infiltration and found that there was less macrophage infiltration in the Pten-null tumors. Using scRNA-seq, they suggest that the observed reduction in macrophages might be due to an immunosuppressive tumor microenvironment.

This mouse model will be of high relevance to the medulloblastoma community, as current models do not reflect the heterogeneity of the disease. In addition, the elegant experimentation into Pten function may be relevant to cancer biologists outside of the medulloblastoma field.

Strengths:

The in-depth characterisation of the mouse model is a major strength of the study, including multiple time points and quantifications. The single-cell sequencing adds a nice molecular feature, and this dataset may be relevant to other researchers with specific questions of Pten function.

Weaknesses:

One weakness of the study was the examination of the macrophage phenotype, which did not include quantification (only single images), so it is difficult to assess whether this reduction of macrophages holds true across multiple samples. Future studies will also be needed to assess whether Pten-mutated patient medulloblastomas also have a differentiation phenotype, but this is difficult to assess given the low number of samples worldwide.

<https://doi.org/10.7554/eLife.108190.1.sa0>